

Vector Locally Repairable Codes With Small Repair Bandwidth and Small Sub-packetization Levels

Jie Li, *Senior Member, IEEE*, Han Cai, *Member, IEEE*, Xiaohu Tang, *Senior Member, IEEE*, Yungshiang S. Han, *Fellow, IEEE*, Bo Bai, *Senior Member, IEEE*, and Gong Zhang

Abstract—Maximum distance separable (MDS) codes in distributed storage systems provide the optimal tradeoff between fault tolerance and storage overhead. As a kind of MDS codes, minimum storage regenerating (MSR) codes have attracted a lot of attention since they are also optimal in terms of repair bandwidth. However, MSR codes suffer from a high repair degree, meaning many helper nodes are needed in the node repair process. Compared to MSR codes, locally repairable codes (LRCs) can significantly reduce the repair degree at the cost of increased storage overhead. The recently introduced concept of vector LRCs combines the advantages of MSR codes and LRCs, providing a tradeoff between repair degree/repair bandwidth and storage overhead. Most existing vector LRCs are built on MSR codes or their shortened versions. However, existing MSR codes have an unavoidably large sub-packetization levels, which also result in large sub-packetization levels in the corresponding vector LRCs. In this paper, we propose a new vector LRC structure, where MDS array codes (without shortening) can be employed as the local codes. Based this new structure, we propose three constructions of vector LRCs with small sub-packetization levels and small repair bandwidth, whose required field sizes are comparable to the code lengths. Additionally, the first two constructions offer a flexible tradeoff between the sub-packetization level and the repair bandwidth, while the third construction has a sub-packetization level of 2, making it easy to implement. Compared to existing vector LRCs, the new vector LRCs provide significantly smaller sub-packetization levels and support a wider range of parameters.

Index Terms—MDS codes, distributed storage, locally repairable codes, repair bandwidth, sub-packetization.

I. INTRODUCTION

DISTRIBUTED storage systems have wide applications in practice, such as Windows Azure Storage [1] and the Hadoop-based systems deployed in Facebook and Yahoo [2]. MDS codes, which provide the optimal tradeoff between fault tolerance and storage overhead, are commonly used in

previous distributed storage systems, e.g., HDFS RAID [3]. However, in general, MDS codes are inefficient when it comes to node repair. In a distributed storage system based on an (n, k) MDS code, repairing a failed node requires contacting any k surviving nodes and downloading the entire content from them. This results in a large *repair degree* and excessive *repair bandwidth*, defined as the number of surviving nodes contacted and the amount of data downloaded to repair a failed node, respectively. Two alternatives have been widely studied in the past decade to enable efficient node repair: regenerating codes [4] and LRCs [5]. The former focuses on minimizing repair bandwidth, while the latter aims at reducing the repair degree.

Two important classes of regenerating codes are MSR and minimum bandwidth regenerating (MBR) codes. The former is obtained by minimizing the storage overhead first and then the repair bandwidth. At the same time, the latter is obtained contrary, i.e., minimizing the repair bandwidth first and then the storage overhead. MSR codes are a type of MDS codes with the minimum repair bandwidth. For example, when encoding a file of size $k\ell$ over $\mathbb{F}_q$ using an (n, k) MSR code with repair degree d ($k \leq d < n$), we obtain n pieces of data, each contains ℓ elements in $\mathbb{F}_q$. When distributing the n pieces of data across a distributed storage system of n nodes, any k nodes can reconstruct the original file. Any failed node can be regenerated by downloading $\frac{d\ell}{d-k+1}$ elements in $\mathbb{F}_q$ from any d helper nodes. Since 2010, MSR codes have attracted a lot of attention [6]–[24].

MSR codes are known for their optimality in terms of repair bandwidth, but they require many helper nodes to be contacted during the node repair process. In [5], the concept of “locality” was introduced to reduce the repair degree. Formally, the i -th code symbol c_i of an (n, k) linear code $\mathcal{C}$ is said to have locality r if c_i can be recovered from at most r other code symbols, where $r \leq k$. It was shown in [5] that the minimum distance $d_{\min}$ of an (n, k) linear code with locality r satisfies

$$d_{\min} \leq n - k - \left\lceil \frac{k}{r} \right\rceil + 2.$$

Codes with locality are also called LRCs in [25]. The concept of locality was later generalized to tolerate more failures in [26]. That is, a failed node can be recovered even in the presence of some additional node failures under a given threshold. Specifically, the i -th code symbol c_i of $\mathcal{C}$ is said to have (r, δ) locality if there exists a punctured code $\mathcal{C}_{S_i}$ such that $i \in S_i \subset \{0, 1, \dots, n-1\}$ with $|S_i| \leq r + \delta - 1$ and the minimum distance of $\mathcal{C}_{S_i}$ is at least δ . Here, $\mathcal{C}_{S_i}$ is obtained by restricting $\mathcal{C}$ to the subset of coordinates corresponding to

The work of H. Cai was supported in part by the National Key R&D Program of China under Grant 2022YFA1005000. The work of Xiaohu Tang was supported in part by the National Natural Science Foundation of China under Grant 62371401. The work of Yungshiang S. Han was supported by the National Key Research and Development Program of China under Grant 2022YFA1004902.

J. Li, B. Bai, and G. Zhang are with the Theory Lab, Central Research Institute, 2012 Labs, Huawei Technologies Co., Ltd., Shatin, New Territories, Hong Kong SAR, China (e-mails: lijie9@huawei.com; baibo8@huawei.com; nicholas.zhang@huawei.com).

H. Cai and X. Tang are with the Information Coding and Transmission Key Lab of Sichuan Province, CSNMT Int. Coop. Res. Centre (MoST), Southwest Jiaotong University, Chengdu, 610031, China (e-mail: hancai@swjtu.edu.cn, xhutang@swjtu.edu.cn).

Yungshiang S. Han is with the Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China, Shenzhen, 518110, China (e-mail: yunghsiangh@gmail.com).

S_i and is usually referred to as local code in the literature. It was shown in [25] that the minimum distance of an (n, k) linear code with (r, δ) locality satisfies

$$d_{\min} \leq n - k + 1 - \left(\left\lceil \frac{k}{r} \right\rceil - 1\right)(\delta - 1). \quad (1)$$

Since then, there has been significant research on the upper bound of the minimum distance of LRCs under various models, as well as the optimal constructions [27]–[37].

A. Related Works

To further reduce the repair bandwidth of LRCs, studies on combining both advantages of regenerating codes and LRCs were considered in [27] and [28], which are referred to as *vector LRCs* in [28]. Intuitively, vector LRCs are constructed by employing regenerating codes or vector codes as the local codes. The main challenge is designing the global parity nodes of vector LRCs to achieve the optimal minimum distance. There are a few works devoted to addressing these issues.

In [27] and [28], upper bounds on the minimum distance were given for general vector LRCs, and a few constructions based on MSR codes or MBR codes achieving the upper bound were also provided. Specifically, in [27], two vector LRC constructions based on generator matrices were proposed. The two constructions were based on MSR codes with large code lengths and arbitrary repair degrees, where the local codes of the vector LRCs are shortened versions of the base MSR codes and some parity nodes of the MSR codes are chosen as the building blocks of the global parity nodes of the vector LRCs. This results in the vector LRCs having sub-packetization levels as large as those of the base MSR codes. In [28], the vector LRCs were obtained by employing MSR codes as the local codes while utilizing a Gabidulin code to encode the global parities; thus, a large finite field is required.

In [38], two vector LRCs based on the MBR codes in [6] and the MSR codes in [12] were proposed. While in [39], vector LRCs based on minimum bandwidth cooperative regeneration codes to repair multiple failed nodes were studied. It is worth noting that when MSR codes are deployed in vector LRCs, they have a large sub-packetization level, as MSR codes are shown to have a sub-packetization level exponential in the code length [40]. This eliminates one of the advantages of LRCs having a sub-packetization level of 1. Set along MSR codes with large code length are deployed to construct local codes in [27] and Gabidulin codes are deployed to build global parity nodes in [28].

Another kind of codes similar to vector LRCs are vector Partial MDS (PMDS) codes, also referred to as PMDS array codes in [41]. They are not only distance-optimal vector LRCs but can also correct any erasure pattern that is information-theoretically correctable. More details can be referred to [41] and the references therein. Similar to vector LRCs, vector PMDS codes also utilize vector codes as local codes, but offer better fault tolerance than vector LRCs. However, this advantage comes with the requirement of large finite fields, e.g., Construction C in [42] requires a finite field with size around $(rn)^{n-r}$. In addition, in [41], explicit constructions

over finite fields with sizes comparable to the code lengths are only provided for codes with two global parity nodes. In contrast, vector LRCs are more flexible and can work for any number of global parity nodes.

B. Main Contributions

This paper focuses on constructing vector LRCs over small finite fields with small sub-packetization levels and small repair bandwidth. Unlike previous constructions in [27], [28], [38], we propose a new structure for vector LRCs by defining the parity-check matrices, where we use MSR codes or MDS array codes with small sub-packetization levels, as opposed to the shortened versions, as the local codes. We also elaborately design the global parities to ensure the minimum distance. This guarantees that the vector LRCs have smaller sub-packetization levels than those in [27] and require smaller finite fields than the one in [28]. Additionally, we characterize the relation between the numbers of local parity nodes and global parity nodes that can achieve the optimal minimum distance.

Based on this new structure, we propose three constructions of vector LRCs. We employ MDS array codes in [43]–[45] along with a newly constructed MDS array code as the local code. The locality of the new vector LRCs is thus guaranteed by these MDS array codes. We elaborately design global parity nodes for each of the new vector LRCs based on the structure of their local codes to achieve the optimal minimum distance.

To summarize, the main contributions of this work are:

- 1) We propose a new structure of vector LRCs by defining the parity-check matrices, and characterizing the relation between the numbers of local parity nodes and global parity nodes that can achieve the optimal minimum distance.
- 2) A new $(r + \delta - 1, r)$ MDS array code with a sub-packetization level of 2 is constructed, which generalizes the one in [41, Construction 3] to support any $\delta \geq 3$. By employing MDS array codes in [43]–[45] and this newly constructed MDS array code, and elaborately design global parity nodes, we construct three new vector LRCs.
- 3) As shown in Table I, the three new vector LRCs have smaller sub-packetization levels than the ones in [27] and [28] since w^m can be smaller than $w^{\lceil \frac{r+\delta-1}{2} \rceil}$ according to $m \mid r + \delta - 1$, and smaller than the one in [38] when $m < \frac{r+\delta-1}{w}$.
- 4) The three new vector LRCs are built on small finite fields with required field sizes comparable to the code lengths.
- 5) The first two new vector LRCs can provide a flexible tradeoff between the sub-packetization level and repair bandwidth.

More extensive comparisons with existing vector LRCs will be carried out in Section VI.

The rest of this paper is organized as follows. Section II introduces some necessary preliminaries. Section III proposes a vector LRC that offers a tradeoff between the sub-packetization level and the repair bandwidth. Section IV introduces a construction of vector LRCs with relatively small

TABLE I
COMPARISON OF THE SUB-PACKETIZATION LEVELS AMONG NEW
DISTANCE OPTIMAL (n, k) VECTOR LRCs AND EXISTING ONES IN [27],
[28], [38], WHERE EACH LOCAL CODE IS AN $(r + \delta - 1, r)$ MDS ARRAY
CODE WITH REPAIR DEGREE $d = r + w - 1$, WHERE $m \mid (r + \delta - 1)$.

	Sub-packetization level ℓ
Construction 5.1 in [27] based on LLT code in [24]	$w^{\lceil \frac{r+\delta+s-1}{2} \rceil}$
Construction 1 in [28] based on LLT code in [24]	$w^{\lceil \frac{r+\delta-1}{2} \rceil}$
Codes with MSR locality [38]	$w^{\frac{r+\delta-1}{w}}$
New vector LRC in Construction 1	$w^{\frac{m}{2}}$
New vector LRC in Construction 2	w^m
New vector LRC in Construction 3	2

repair bandwidth and field size. A new vector LRC with a sub-packetization level of 2 is presented in Section V. Section VI extensively compares the new vector LRCs and existing ones. Finally, Section VII concludes the paper.

II. PRELIMINARIES

This section introduces some preliminaries of vector codes, vector LRCs, and a series of special partitions for a given basis set. Throughout this paper, denote by q a prime power and $\mathbf{F}_q$ the finite field with q elements. For any two integers a and b with $b > a$, denote by $[a : b]$ the set $\{a, a + 1, \dots, b - 1\}$.

A. Vector Codes

Definition 1. ([27]) An $\mathbf{F}_q$ -linear vector code (or array code) of block length n is a code $\mathcal{C}$ such that

- i) For a codeword $(\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_{n-1}) \in \mathcal{C}$, $\mathbf{c}_i \in \mathbf{F}_q^\ell$ for all $i \in [0 : n)$, where $\ell \geq 1$ and is usually referred to as sub-packetization level,
- ii) For given $\mathbf{c}, \mathbf{c}' \in \mathcal{C}$ and $x, y \in \mathbf{F}_q$,

$$x\mathbf{c} + y\mathbf{c}' = (x\mathbf{c}_0 + y\mathbf{c}'_0, x\mathbf{c}_1 + y\mathbf{c}'_1, \dots, x\mathbf{c}_{n-1} + y\mathbf{c}'_{n-1}) \in \mathcal{C},$$

$$\text{where } \mathbf{c} = (\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_{n-1}), \mathbf{c}' = (\mathbf{c}'_0, \mathbf{c}'_1, \dots, \mathbf{c}'_{n-1}).$$

A vector code $\mathcal{C}$ can be associated with an $\mathbf{F}_q$ -linear scalar code $\mathcal{C}^{(s)}$ of length $N = n\ell$. This is done by expanding each codeword $(\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_{n-1})$ in $\mathcal{C}$ into a vector in $\mathbf{F}_q^{N}$, where $\mathbf{c}_i$ corresponds to the $i\ell, i\ell + 1, \dots, (i\ell + \ell - 1)$ -th symbols in the vector. For simplicity, in this paper, we only consider the vector code where the associated scalar code has dimension $k\ell$ for some $k \in [1 : n]$. Such vector codes are referred to as (n, k) vector codes with sub-packetization level ℓ , which can be defined by a parity-check matrix of the following form

$$H = \begin{pmatrix} A_{0,0} & A_{0,1} & \cdots & A_{0,n-1} \\ A_{1,0} & A_{1,1} & \cdots & A_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n-k-1,0} & A_{n-k-1,1} & \cdots & A_{n-k-1,n-1} \end{pmatrix}, \quad (2)$$

where $A_{i,j}$ is an $\ell \times \ell$ matrix over $\mathbf{F}_q$. For convenience, we refer to

$$H[J] = (A_{0,j}^\top, A_{1,j}^\top, \dots, A_{n-k-1,j}^\top)^\top$$

as the j -th block column of H , and $H[J]$ represents the sub-matrix of H formed by the block columns with indices indicated by $J \subset [0 : n)$.

Lemma 1. ([15]) For an (n, k) vector code with sub-packetization level ℓ admitting the parity-check matrix in (2), it is an MDS array code if and only if any $(n - k) \times (n - k)$ sub-block matrix $H[J]$ of H is non-singular, where $J \subset [0 : n)$ and $|J| = n - k$.

In particular, if $A_{t,i} = A_i^t$, $t \in [0 : n - k)$, $i \in [0 : n)$ for some matrices A_i of order ℓ , we have the following result to verify the non-singularity of the block matrices.

Lemma 2. (Block Vandermonde matrix, [15, Lemma 14]) For $u > 1$, let $B_0, \dots, B_{u-1}$ be $\ell \times \ell$ matrices such that $B_i B_j = B_j B_i$ and $B_i - B_j$ is non-singular for all $i, j \in [0 : u)$ with $i \neq j$, then the matrix

$$\begin{pmatrix} I_\ell & I_\ell & \cdots & I_\ell \\ B_0 & B_1 & \cdots & B_{u-1} \\ \vdots & \vdots & \ddots & \vdots \\ B_0^{u-1} & B_1^{u-1} & \cdots & B_{u-1}^{u-1} \end{pmatrix}$$

is non-singular, where I_ℓ denotes the identity matrix of order ℓ , and will be abbreviated as I in the sequel if the context is clear.

Lemma 3. ([43, Lemma 1]) For $u > 1$, $t, i \in [0 : u)$, let $B_{t,i}$ be an $\ell \times \ell$ upper triangular matrix, i.e.,

$$B_{t,i}(a, b) = 0 \text{ for } 0 \leq b < a < \ell, \quad (3)$$

where $B_{t,i}(a, b)$ denotes the entry in column a and row b of the matrix $B_{t,i}$, then the block matrix

$$B = \begin{pmatrix} B_{0,0} & B_{0,1} & \cdots & B_{0,u-1} \\ B_{1,0} & B_{1,1} & \cdots & B_{1,u-1} \\ \vdots & \vdots & \ddots & \vdots \\ B_{u-1,0} & B_{u-1,1} & \cdots & B_{u-1,u-1} \end{pmatrix}$$

is non-singular if

- i) $B_{t,i}(a, a) = (B_{1,i}(a, a))^t$ for $i, t \in [0 : u)$ and $a \in [0 : \ell)$,
- ii) $B_{1,i}(a, a) \neq B_{1,j}(a, a)$ for any $i, j \in [0 : u)$ with $j \neq i$ and $a \in [0 : \ell)$.

B. Vector Locally Repairable Codes

Definition 2. ([27]) Let $\mathcal{C}$ be an (n, k) linear vector code with sub-packetization ℓ over $\mathbf{F}_q$, the i -th code symbol c_i of $\mathcal{C}$ is said to have (r, δ) locality if there exists a punctured vector code $\mathcal{C}_S$ such that $i \in S \subset [0 : n)$ with $|S| \leq r + \delta - 1$ and the minimum distance of $\mathcal{C}_S$ is at least δ . The code $\mathcal{C}$ is said to have an all-symbol locality if all code symbols have (r, δ) locality.

A vector LRC will be reduced to an LRC if $\ell = 1$.

In particular, if there exists pairwise disjoint subsets $S_0, S_1, \dots, S_{\mu-1}$ of $[0 : n)$ such that

$$S_0 \cup S_1 \cup \cdots \cup S_{\mu-1} = [0 : n), \quad (4)$$

$|S_j| \leq r + \delta - 1$ and $\mathcal{C}_{S_j}$ has minimum distance at least δ for all $j \in [0 : \mu)$, then we say that $\mathcal{C}$ is a vector LRC with disjoint local groups, where each S_j is referred to as a local group and $\mathcal{C}_{S_j}$ is called a local code. Furthermore, $\mathcal{C}$ is said

to have *uniform* (r, δ) all-symbol locality if $\mathcal{C}$ is a vector LRC with disjoint local groups and

$$|S_j| = r + \delta - 1 \text{ for all } j \in [0 : \mu). \quad (5)$$

In this paper, we consider (n, k) vector LRCs with uniform (r, δ) all-symbol locality, and an MDS array code encodes each local group. In [27] and [28], it was shown that the minimum distance of such vector LRCs satisfies

$$d_{\min} \leq n - k + 1 - \left(\left\lceil \frac{k}{r} \right\rceil - 1 \right) (\delta - 1), \quad (6)$$

which coincides with that of scalar LRCs.

Let ℓ denote the sub-packetization level and μ the number of local groups, where $\mu = \frac{n}{r+\delta-1}$ is an integer by (4) and (5). Assume that $S_i = [(r+\delta-1)i : (r+\delta-1)(i+1))$ is the i -th local group, i.e., $\mathcal{C}_{S_i}$ is an $(r+\delta-1, r)$ MDS array code with sub-packetization level ℓ , where $i \in [0 : \mu)$. Immediately, every such $(n, k = \mu r - s)$ vector LRC permits a form of the following parity-check matrix

$$\mathcal{H} = \begin{pmatrix} H_0 & & & \\ & H_1 & & \\ & & \ddots & \\ & & & H_{\mu-1} \\ P_0 & P_1 & \cdots & P_{\mu-1} \end{pmatrix}, \quad (7)$$

where H_i is a $(\delta-1)\ell \times (r+\delta-1)\ell$ matrix, e.g.,

$$H_i = \begin{pmatrix} A_{0,0}^{(i)} & A_{0,1}^{(i)} & \cdots & A_{0,r+\delta-2}^{(i)} \\ A_{1,0}^{(i)} & A_{1,1}^{(i)} & \cdots & A_{1,r+\delta-2}^{(i)} \\ \vdots & \vdots & \ddots & \vdots \\ A_{\delta-2,0}^{(i)} & A_{\delta-2,1}^{(i)} & \cdots & A_{\delta-2,r+\delta-2}^{(i)} \end{pmatrix}, \quad (8)$$

P_i is an $s\ell \times (r+\delta-1)\ell$ matrix, e.g.,

$$P_i = \begin{pmatrix} A_{\delta-1,0}^{(i)} & A_{\delta-1,1}^{(i)} & \cdots & A_{\delta-1,r+\delta-2}^{(i)} \\ A_{\delta,0}^{(i)} & A_{\delta,1}^{(i)} & \cdots & A_{\delta,r+\delta-2}^{(i)} \\ \vdots & \vdots & \ddots & \vdots \\ A_{\delta+s-2,0}^{(i)} & A_{\delta+s-2,1}^{(i)} & \cdots & A_{\delta+s-2,r+\delta-2}^{(i)} \end{pmatrix}, \quad (9)$$

$A_{t,j}^{(i)}$ is a square matrix of order ℓ , and s denotes the number of global parities, respectively.

Lemma 4. For an (n, k) vector LRC defined in the form of (7) with sub-packetization level ℓ and (r, δ) all-symbol locality, it is optimal w.r.t. the bound in (6) if $s < r$ and

$$\text{rank} \left(\begin{pmatrix} H_{i_0}[T_0] & H_{i_1}[T_1] & \cdots & H_{i_{b-1}}[T_{b-1}] \\ P_{i_0}[T_0] & P_{i_1}[T_1] & \cdots & P_{i_{b-1}}[T_{b-1}] \end{pmatrix} \right) = \ell(\delta + s - 1) \quad (10)$$

for any $0 \leq i_0 < i_1 < \cdots < i_{b-1} < \mu$, $T_i \subset [0 : r+\delta-1)$ and $|T_i| > \delta-1$ for $0 \leq i < b$ such that $|T_0| + |T_1| + \cdots + |T_{b-1}| = \delta + s - 1$, where b is a positive integer.

Proof. Plug $k = \mu r - s$ in (6), we have

$$\begin{aligned} d_{\min} &\leq n - (\mu r - s) + 1 - \left(\left\lceil \frac{\mu r - s}{r} \right\rceil - 1 \right) (\delta - 1) \\ &= \mu(\delta - 1) + s + 1 - \left(\left\lceil \frac{\mu r - s}{r} \right\rceil - 1 \right) (\delta - 1) \\ &= \delta + s, \end{aligned}$$

where the last equality follows from $s < r$. That is, given $s < r$, a vector LRC defined in the form of (7) is optimal w.r.t. the bound in (6) if it can tolerate any $\delta + s - 1$ node failures.

Suppose that the $\delta + s - 1$ failed nodes are dispersed across b different local groups with each group containing more than $\delta - 1$ failed nodes, say groups $i_0, i_1, \dots, i_{b-1}$, where b is a positive integer and $0 \leq i_0 < i_1 < \cdots < i_{b-1} < \mu$. Assume that the set of indices of the failed nodes in group i_j is T_j for $0 \leq j < b$, i.e., $|T_j| > \delta - 1$ and $|T_0| + |T_1| + \cdots + |T_{b-1}| = \delta + s - 1$, then the original file can be reconstructed if the rank of the following matrix

$$\hat{\mathcal{H}} = \begin{pmatrix} H_{i_0}[T_0] & & & \\ & H_{i_1}[T_1] & & \\ & & \ddots & \\ & & & H_{i_{b-1}}[T_{b-1}] \\ P_{i_0}[T_0] & P_{i_1}[T_1] & \cdots & P_{i_{b-1}}[T_{b-1}] \end{pmatrix}$$

is at least $\ell(\delta + s - 1)$. Note that $\hat{\mathcal{H}}$ is equivalent to

$$Q = \begin{pmatrix} H_{i_0}[T_0] & & & \\ & H_{i_1}[T_1] & & \\ & & \ddots & \\ & & & H_{i_{b-1}}[T_{b-1}] \\ H_{i_0}[T_0] & H_{i_1}[T_1] & \cdots & H_{i_{b-1}}[T_{b-1}] \\ P_{i_0}[T_0] & P_{i_1}[T_1] & \cdots & P_{i_{b-1}}[T_{b-1}] \end{pmatrix}$$

under elementary row transformation, and

$$\begin{aligned} \text{rank}(\hat{\mathcal{H}}) &= \text{rank}(Q) \\ &\geq \text{rank} \left(\begin{pmatrix} H_{i_0}[T_0] & H_{i_1}[T_1] & \cdots & H_{i_{b-1}}[T_{b-1}] \\ P_{i_0}[T_0] & P_{i_1}[T_1] & \cdots & P_{i_{b-1}}[T_{b-1}] \end{pmatrix} \right), \end{aligned}$$

then we obtain the desired result. $\square$

C. Partition of Basis $\{e_0, \dots, e_{\ell-1}\}$

Let $h = w^m$ for two integers w and m with $w, m \geq 2$, and let $e_0, \dots, e_{w^m-1}$ be a basis of $\mathbf{F}_q^{w^m}$. For example, they can be set as the standard basis, i.e.,

$$e_i = (0, \dots, 0, 1, 0, \dots, 0), \quad i \in [0 : w^m),$$

with only the i -th entry being nonzero.

In [11] and [44], a series of special partitions of the set $\{e_0, \dots, e_{w^m-1}\}$ was given. In what follows, we revisit these partitions as they play a crucial role in Construction 2 in Section IV. We follow the notations in [11] and [44] hereafter for consistency. Given an integer $0 \leq b < w^m$, denote by $(b_0, \dots, b_{m-1})$ its w -ary expansion, i.e., $b = \sum_{j=0}^{m-1} w^{m-1-j} b_j$. For $0 \leq i < m$ and $0 \leq t < w$, define a subset of $\{e_0, \dots, e_{w^m-1}\}$ as

$$V_{i,t} = \{e_b | b_i = t, 0 \leq b < w^m\}, \quad (11)$$

where b_i is the i -th element in the w -ary expansion of b .

For any $i \in [0 : m)$ and $t \in [0 : w)$, it is straightforward that $\{V_{i,0}, V_{i,1}, \dots, V_{i,w-1}\}$ is a partition of the set $\{e_0, \dots, e_{w^m-1}\}$ and $|V_{i,t}| = w^{m-1}$. Table II gives two examples of the set partitions defined in (11).

TABLE II

(A) AND (B) DENOTE THE m PARTITIONS OF THE SET $\{e_0, \dots, e_{w^m-1}\}$ DEFINED BY (11) FOR $m = 3, w = 2$, AND $m = 2, w = 3$, RESPECTIVELY.

i	0	1	2	i	0	1	2
$V_{i,0}$	e_0	e_0	e_0	$V_{i,1}$	e_4	e_2	e_1
	e_1	e_1	e_2		e_5	e_3	e_3
	e_2	e_4	e_4		e_6	e_6	e_5
	e_3	e_5	e_6		e_7	e_7	e_7

(A)

i	0	1	i	0	1	i	0	1
$V_{i,0}$	e_0	e_0	$V_{i,1}$	e_3	e_1	$V_{i,2}$	e_6	e_2
	e_1	e_3		e_4	e_4		e_7	e_5
	e_2	e_6		e_5	e_7		e_8	e_8

(B)

Based on the m set partitions defined by (11), let us define

$$V_{i+sm,t} = V_{i,t}, \quad (12)$$

where $i \in [0 : m)$, $s \geq 1$, and $t \in [0 : w)$.

For convenience of notation, we also denote by $V_{i,t}$ the $w^{m-1} \times w^m$ matrix, whose rows are formed by vectors e_i in their corresponding sets, respectively, such that i is sorted in ascending order. For example, when $w = 2$ and $m = 3$, $V_{1,0}$ can be viewed as a 4×8 matrix as follows

$$V_{1,0} = (e_0^\top \ e_1^\top \ e_4^\top \ e_5^\top)^\top.$$

III. A NEW CONSTRUCTION OF VECTOR LRCs WITH SMALL SUB-PACKETIZATION LEVEL

In this section, we present a construction of vector LRCs, which yields vector LRCs with a small sub-packetization level and small repair bandwidth.

Construction 1. For $s \geq 1$, $\delta \geq 3$, $m > 1$ such that $2 \mid m$, and $1 < w \leq \delta - 1$, let $\ell = w^{\frac{m}{2}}$, define $\bar{A}_{t,j'}$ as in (13) where $\lambda_{j',0}, \lambda_{j',1}, \dots, \lambda_{j',w-1} \in \mathbf{F}_q \setminus \{0\}$. Let $\mu > 1$ and $r + \delta - 1 \geq m$, we construct an $(n = \mu(r + \delta - 1), k = \mu r - s)$ vector LRC over $\mathbf{F}_q$ with the parity-check matrix having the form as in (7)–(9) and sub-packetization level ℓ , where

$$A_{t,j}^{(i)} = y_i^t x_j^t \bar{A}_{t,j \% m}, t \in [0 : \delta + s - 1), j \in [0 : r + \delta - 1), \quad (14)$$

$j \% m$ denotes the integer $j \bmod m$, and $x_j, y_i \in \mathbf{F}_q \setminus \{0\}$. For convenience, we assume that $m \mid r + \delta - 1$. Note that the parameter m controls the tradeoff between the sub-packetization level and the repair bandwidth.

In Construction 1, we utilize the MDS array code $\mathcal{C}_2$ in [43] if $w = \delta - 1$ and $\mathcal{C}_1$ in [45] if $w < \delta - 1$ as the local codes. Hence, H_i in (7) precisely represents the parity-check matrix of the MDS array code either $\mathcal{C}_2$ in [43] or $\mathcal{C}_1$ in [45]. For the global parity checks in (9), we elaborately design each block in (9) as a diagonal matrix, where the diagonal entries are powers of the corresponding diagonal entries of the blocks in H_i . This arrangement ensures that the matrix formed by

vertically concatenating H_i and P_i constitutes a block upper triangular matrix, facilitating the verification of the minimum distance using Lemma 3.

Lemma 5. Let $\gamma^* = \frac{d\ell}{w}$ denote the repair bandwidth of a $(r + \delta - 1, r)$ MSR code with sub-packetization level ℓ and repair degree d , then each local code of the vector LRC in Construction 1 is an $(r + \delta - 1, r)$ MDS array code with the repair bandwidth being

$$\gamma = \left(1 + \frac{(\frac{r+\delta-1}{m} - 1)(w-1)}{d}\right) \gamma^*$$

and repair degree $d = r + w - 1$ if

- i) $x_l \lambda_{l \% m, u} \neq x_j \lambda_{j \% m, u'}$ for $u, u' \in [0 : w)$ and $l, j \in [0 : r + \delta - 1)$ with $l \not\equiv j \bmod \frac{m}{2}$,
- ii) $x_l \lambda_{l \% m, u} \neq x_j \lambda_{j \% m, u}$ for $u \in [0 : w)$ and $l, j \in [0 : r + \delta - 1)$ with $l \neq j$ and $l \equiv j \bmod \frac{m}{2}$,
- iii) $\lambda_{j,u} \neq \lambda_{j,u'}$ for $u, u' \in [0 : w)$ with $u \neq u'$ and $j \in [0 : m)$,

hold, and additionally

- iv) $x_l \lambda_{l \% m, 0} \neq x_j \lambda_{j \% m, u}$ holds for $u \in [1 : w)$ and $l, j \in [0 : r + \delta - 1)$ with $l \equiv j \bmod \frac{m}{2}$ and $l \not\equiv j \bmod m$,

holds if $w < \delta - 1$.

Proof. Note that H_i is the parity-check matrix of the MDS array code $\mathcal{C}_2$ in [43] and $\mathcal{C}_1$ in [45] if $w = \delta - 1$ and $w < \delta - 1$, respectively, thus from the proofs of Theorem 4 in [43] and Theorem 3 in [45], we have the desired result. $\square$

Theorem 1. Based on the conditions in Lemma 5, the vector LRC in Construction 1 is optimal w.r.t. the bound in (6) if $s < r$ and

$$\begin{aligned} & \{y_{i_0} x_j \lambda_{j \% m, u} \mid j \in [0 : r + \delta - 1), u \in [0 : w)\} \\ & \cap \{y_{i_1} x_j \lambda_{j \% m, u} \mid j \in [0 : r + \delta - 1), u \in [0 : w)\} = \emptyset \end{aligned} \quad (15)$$

for $0 \leq i_0 < i_1 < \mu$, where $y_{i_0} x_j \lambda_{j \% m, u}$ is an entry on the diagonal of $A_{1,j}^{(i_0)}$ according to (13) and (14).

Remark 1. Intuitively, (15) implies that the diagonal entries in the matrices $A_{1,j}^{(i_0)}$, $j \in [0 : r + \delta - 1)$ (associated with local group i_0) are distinct from those in $A_{1,j}^{(i_1)}$, $j \in [0 : r + \delta - 1)$ (associated with local group i_1), where $0 \leq i_0 < i_1 < \mu$.

Proof. If the conditions in Lemma 5 are satisfied, then each local code of the vector LRC in Construction 1 is a $(r + \delta - 1, r)$ MDS array code, and thus the LRC in Construction 1 has the (r, δ) locality. Thus we only need to verify the requirement in Lemma 4.

For any $0 \leq i_0 < i_1 < \dots < i_{b-1} < \mu$, $T_i \subset [0 : r + \delta - 1)$ and $|T_i| > \delta - 1$ for $0 \leq i < b$ such that $|T_0| + |T_1| + \dots + |T_{b-1}| = \delta + s - 1$. Let $t_i = |T_i|$ and $T_i = \{j_0^{(i)}, \dots, j_{t_i-1}^{(i)}\}$ for $i \in [0 : b)$. Define

$$Q = \begin{pmatrix} H_{i_0}[T_0] & H_{i_1}[T_1] & \dots & H_{i_{b-1}}[T_{b-1}] \\ P_{i_0}[T_0] & P_{i_1}[T_1] & \dots & P_{i_{b-1}}[T_{b-1}] \end{pmatrix}.$$

By substituting (13) and (14) into Q , we derive (16) on the next page. In the following, we prove that Q is non-singular.

- i) By (13) and (14), we observe that each block entry $A_{t,j}^{(i_u)}$ in Q is upper triangular since $\bar{A}_{t,j \% m}$ in (13)

$$\bar{A}_{t,j'} = \begin{cases} \sum_{a=0}^{\ell-1} \lambda_{j',a_{j'}}^t e_a^\top e_a + \sum_{a=0, a_{j'}=0}^{\ell-1} \sum_{u=1}^{w-1} (\lambda_{j',0}^t - \lambda_{j',u}^t) e_a^\top e_{a(j',u)}, & \text{if } j' \in [0 : \frac{m}{2}), t \in [0 : \delta - 1), \\ \sum_{a=0}^{\ell-1} \lambda_{j',a_{j'}}^t e_a^\top e_a, & \text{if } j' \in [0 : \frac{m}{2}), t \in [\delta - 1 : \delta + s - 1), \\ \sum_{a=0}^{\ell-1} \lambda_{j',a_{j'} - \frac{m}{2}}^t e_a^\top e_a, & \text{if } j' \in [\frac{m}{2} : m), t \in [0 : \delta + s - 1). \end{cases} \quad (13)$$

$$Q = \begin{pmatrix} H_{i_0}[T_0] & H_{i_1}[T_1] & \cdots & H_{i_{b-1}}[T_{b-1}] \\ P_{i_0}[T_0] & P_{i_1}[T_1] & \cdots & P_{i_{b-1}}[T_{b-1}] \end{pmatrix} = \begin{pmatrix} A_{0,j_0^{(0)}}^{(i_0)} & \cdots & A_{0,j_{t_0-1}^{(0)}}^{(i_0)} & \cdots & A_{0,j_0^{(b-1)}}^{(i_{b-1})} & \cdots & A_{0,j_{t_{b-1}-1}^{(b-1)}}^{(i_{b-1})} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ A_{\delta-2,j_0^{(0)}}^{(i_0)} & \cdots & A_{\delta-2,j_{t_0-1}^{(0)}}^{(i_0)} & \cdots & A_{\delta-2,j_0^{(b-1)}}^{(i_{b-1})} & \cdots & A_{\delta-2,j_{t_{b-1}-1}^{(b-1)}}^{(i_{b-1})} \\ A_{\delta-1,j_0^{(0)}}^{(i_0)} & \cdots & A_{\delta-1,j_{t_0-1}^{(0)}}^{(i_0)} & \cdots & A_{\delta-1,j_0^{(b-1)}}^{(i_{b-1})} & \cdots & A_{\delta-1,j_{t_{b-1}-1}^{(b-1)}}^{(i_{b-1})} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ A_{\delta+s-2,j_0^{(0)}}^{(i_0)} & \cdots & A_{\delta+s-2,j_{t_0-1}^{(0)}}^{(i_0)} & \cdots & A_{\delta+s-2,j_0^{(b-1)}}^{(i_{b-1})} & \cdots & A_{\delta+s-2,j_{t_{b-1}-1}^{(b-1)}}^{(i_{b-1})} \end{pmatrix} \quad (16)$$

is upper triangular according to the proof of Theorem 1 in [43].

ii) By (13) and (14), for $t \in [0 : \delta + s - 1)$, $u \in [0 : b)$, $v \in [0 : t_u)$, and $a \in [0 : \ell)$, we have

$$A_{t,j_v^{(u)}}^{(i_u)}(a, a) = \left(A_{1,j_v^{(u)}}^{(i_u)}(a, a) \right)^t.$$

iii) For $a \in [0 : \ell)$, $u, g \in [0 : b)$, $v \in [0 : t_u)$, and $z \in [0 : t_g)$ with $(u, v) \neq (g, z)$, we have

$$\begin{aligned} A_{1,j_v^{(u)}}^{(i_u)}(a, a) &= y_{i_u} x_{j_v^{(u)}} \bar{A}_{1,j_v^{(u)}}^{(i_u)}(a, a) \\ &= y_{i_u} x_{j_v^{(u)}} \lambda_{j_v^{(u)} \% m, a_{j_v^{(u)} \% \frac{m}{2}}} \\ &\neq y_{i_g} x_{j_z^{(g)}} \lambda_{j_z^{(g)} \% m, a_{j_z^{(g)} \% \frac{m}{2}}} \\ &= A_{1,j_z^{(g)}}^{(i_g)}(a, a), \end{aligned}$$

where the first and second equalities follow from (14) and (13), respectively, the inequality follows from i) and ii) of Lemma 5 if $u = g$ and from (15) if $u \neq g$.

Applying Lemma 3, we have that the matrix Q is non-singular.

This completes the proof according to Lemma 4. $\square$

Theorem 2. *The requirements in Lemma 5 and (15) can be satisfied if q is a prime power such that*

$$q > \begin{cases} n(w+2)/2, & \text{if } w = 2, \\ n(w+1)/2, & \text{if } w \in [3 : \delta - 1), \\ nw/2, & \text{if } w = \delta - 1 > 2. \end{cases} \quad (17)$$

Proof. Let c be a primitive element in $\mathbf{F}_q$ with q satisfying (17). For $i \in [0 : \mu)$, $j \in [0 : r + \delta - 1)$, $j' \in [0 : \frac{m}{2})$, $g \in \{0, 1\}$, and $u \in [0 : w)$,

• If $w = 2$, set

$$\begin{aligned} y_i &= c^{i(r+\delta-1)(w+2)/2}, \quad x_j = c^{\lfloor \frac{j}{m} \rfloor \frac{m}{2}(w+2)}, \\ \lambda_{\frac{m}{2}+j',u} &= c^{j'(w+2)+u+gw}, \end{aligned} \quad (18)$$

• If $w \in [3 : \delta - 1)$, set

$$\begin{aligned} y_i &= c^{i(r+\delta-1)(w+1)/2}, \quad x_j = c^{\lfloor \frac{j}{m} \rfloor \frac{m}{2}(w+1)}, \\ \lambda_{j',u} &= c^{j'(w+1)+u}, \quad \lambda_{\frac{m}{2}+j',0} = c^{j'(w+1)+w}, \end{aligned}$$

and $\lambda_{\frac{m}{2}+j',u} = c^{j'(w+1)+u \% (w-1)+1}$ for $u \geq 1$;

• If $w = \delta - 1 > 2$, set

$$\begin{aligned} y_i &= c^{i(r+\delta-1)w/2}, \quad x_j = c^{\lfloor \frac{j}{m} \rfloor \frac{m}{2}w}, \\ \lambda_{g\frac{m}{2}+j',u} &= c^{j'(w+u+g) \% w}. \end{aligned}$$

The proof is similar to that of Theorem 5 in [43] when $w = \delta - 1$ and Theorem 4 in [45] for $w \geq 3$, thus we omit it here. In the following, we verify that the requirements in Lemma 5 and (15) are satisfied for $w = 2$.

Let $u, u' \in [0 : w)$ and $l, j \in [0 : r + \delta - 1)$, and we rewrite $l \% m$ and $j \% m$ as

$$l \% m = \frac{m}{2}g_0 + l' \quad \text{and} \quad j \% m = \frac{m}{2}g_1 + j',$$

respectively, where $g_0, g_1 = 0, 1$, and $l', j' \in [0 : \frac{m}{2})$.

i) By (18), we have

$$\begin{aligned} & \frac{x_l \lambda_{l \% m, u}}{x_j \lambda_{j \% m, u'}} \\ &= \frac{c^{\lfloor \frac{l}{m} \rfloor \frac{m}{2}(w+2)} c^{l'(w+2)+u+g_0 w}}{c^{\lfloor \frac{j}{m} \rfloor \frac{m}{2}(w+2)} c^{j'(w+2)+u'+g_1 w}} \\ &= c^{(\lfloor \frac{l}{m} \rfloor - \lfloor \frac{j}{m} \rfloor) \frac{m}{2}(w+2) + (l' - j')(w+2) + u - u' + (g_0 - g_1)w} \\ &\neq 1, \end{aligned}$$

$$\begin{aligned}
0 &\neq \left| \left(\left\lfloor \frac{l}{m} \right\rfloor - \left\lfloor \frac{j}{m} \right\rfloor \right) \frac{m}{2}(w+2) + (l' - j')(w+2) + u - u' + (g_0 - g_1)w \right| \\
&\leq \left\lfloor \frac{r+\delta-2}{m} \right\rfloor \frac{m}{2}(w+2) + \left(\frac{m}{2} - 1 \right)(w+2) + w - 1 + w \\
&= \frac{(r+\delta-1)(w+2)}{2} + w - 3 \\
&= \frac{n(w+2)}{2\mu} + w - 3 \\
&< q - 1
\end{aligned} \tag{19}$$

where the inequality follows from (19) on the next page. Therefore, i) and ii) of Lemma 5 are satisfied.

- ii) For $u, u' \in [0 : w)$ with $u \neq u'$ and $j \in [0 : m)$, it is obvious that $\lambda_{j,u} \neq \lambda_{j,u'}$ by (18). That is, iii) of Lemma 5 is fulfilled.
- iii) Now let us verify (15). For $0 \leq i_0 < i_1 < \mu$, by (18), we have

$$\begin{aligned}
&\frac{y_{i_0} x_l \lambda_{l \% m, u}}{y_{i_1} x_j \lambda_{j \% m, u'}} \\
&= \frac{c^{i_0(r+\delta-1)(w+2)/2} c^{\lfloor \frac{l}{m} \rfloor \frac{m}{2}(w+2)} c^{l'(w+2)+u+g_0 w}}{c^{i_1(r+\delta-1)(w+2)/2} c^{\lfloor \frac{j}{m} \rfloor \frac{m}{2}(w+2)} c^{j'(w+2)+u'+g_1 w}} \\
&= c^{(i_0-i_1)(r+\delta-1)(w+2)/2 + (\lfloor \frac{l}{m} \rfloor - \lfloor \frac{j}{m} \rfloor) \frac{m}{2}(w+2)} \\
&\quad \cdot c^{(l'-j')(w+2)+u-u'+(g_0-g_1)w} \\
&\neq 1,
\end{aligned} \tag{20}$$

where the inequality follows from (21) on the next page. $\square$

Example 1. We present an example of an $(n = 16, k = 10)$ vector LRC in Construction 1. Let $r = 6, \delta = 3, w = 2, m = 4, \mu = 2$, and $s = 2$. Then, the matrices $\bar{A}_{t,j}$, $t, j \in [0 : 4)$ in (13) are as follows:

$$\begin{aligned}
\bar{A}_{t,0} &= \begin{pmatrix} \lambda_{0,0}^t & & & \\ & \lambda_{0,0}^t - \lambda_{0,1}^t & & \\ & & \lambda_{0,0}^t - \lambda_{0,1}^t & \\ & & & \lambda_{0,1}^t \end{pmatrix}, \quad t \in [0, 2), \\
\bar{A}_{t,1} &= \begin{pmatrix} \lambda_{1,0}^t & \lambda_{1,0}^t - \lambda_{1,1}^t & & \\ & \lambda_{1,1}^t & & \\ & & \lambda_{1,0}^t & \lambda_{1,0}^t - \lambda_{1,1}^t \\ & & & \lambda_{1,1}^t \end{pmatrix}, \quad t \in [0, 2),
\end{aligned}$$

$$\begin{aligned}
\bar{A}_{t,0} &= \begin{pmatrix} \lambda_{0,0}^t & & & \\ & \lambda_{0,0}^t & & \\ & & \lambda_{0,1}^t & \\ & & & \lambda_{0,1}^t \end{pmatrix}, \quad t \in [2, 4), \\
\bar{A}_{t,1} &= \begin{pmatrix} \lambda_{1,0}^t & & & \\ & \lambda_{1,1}^t & & \\ & & \lambda_{1,0}^t & \\ & & & \lambda_{1,1}^t \end{pmatrix}, \quad t \in [2, 4),
\end{aligned}$$

$$\begin{aligned}
\bar{A}_{t,2} &= \begin{pmatrix} \lambda_{2,0}^t & & & \\ & \lambda_{2,0}^t & & \\ & & \lambda_{2,1}^t & \\ & & & \lambda_{2,1}^t \end{pmatrix}, \quad t \in [0, 4), \\
\bar{A}_{t,3} &= \begin{pmatrix} \lambda_{3,0}^t & & & \\ & \lambda_{3,1}^t & & \\ & & \lambda_{3,0}^t & \\ & & & \lambda_{3,1}^t \end{pmatrix}, \quad t \in [0, 4).
\end{aligned}$$

$$A_{t,j}^{(i)} = y_i^t x_j^t \bar{A}_{t,j \% 4}, \quad i \in [0 : 2), j \in [0 : 8), t \in [0 : 4).$$

Consider the scenario in which we need to reconstruct the original file by connecting nodes excluding nodes 0, 2, 8, and 15, i.e., we need to verify the non-singularity of the following matrix

$$Q = \begin{pmatrix} H_0[\{0, 2\}] & H_1[\{0, 7\}] \\ P_0[\{0, 2\}] & P_1[\{0, 7\}] \end{pmatrix}$$

according to Lemma 4. Note that

$$\begin{aligned}
Q &= \begin{pmatrix} A_{0,0}^{(0)} & A_{0,2}^{(0)} & A_{0,0}^{(1)} & A_{0,7}^{(1)} \\ A_{1,0}^{(0)} & A_{1,2}^{(0)} & A_{1,0}^{(1)} & A_{1,7}^{(1)} \\ A_{2,0}^{(0)} & A_{2,2}^{(0)} & A_{2,0}^{(1)} & A_{2,7}^{(1)} \\ A_{3,0}^{(0)} & A_{3,2}^{(0)} & A_{3,0}^{(1)} & A_{3,7}^{(1)} \end{pmatrix} \\
&= \begin{pmatrix} I & I & I & I \\ y_0 x_0 \bar{A}_{1,0} & y_0 x_2 \bar{A}_{1,2} & y_1 x_0 \bar{A}_{1,0} & y_1 x_7 \bar{A}_{1,3} \\ y_0^2 x_0^2 \bar{A}_{2,0} & y_0^2 x_2^2 \bar{A}_{2,2} & y_1^2 x_0^2 \bar{A}_{2,0} & y_1^2 x_7^2 \bar{A}_{2,3} \\ y_0^3 x_0^3 \bar{A}_{3,0} & y_0^3 x_2^3 \bar{A}_{3,2} & y_1^3 x_0^3 \bar{A}_{3,0} & y_1^3 x_7^3 \bar{A}_{3,3} \end{pmatrix}.
\end{aligned}$$

It is evident that each block in block matrix Q is an upper-triangular matrix, and i) of Lemma 3 holds for the matrix Q . If $A_{1,0}^{(0)}(a, a), A_{1,2}^{(0)}(a, a), A_{1,0}^{(1)}(a, a)$, and $A_{1,7}^{(1)}(a, a)$ are pairwise distinct for all $a \in [0 : 4)$, i.e., ii) of Lemma 3 holds, then Q is non-singular according to Lemma 3. Below, we verify that $A_{1,0}^{(0)}(a, a), A_{1,2}^{(0)}(a, a), A_{1,0}^{(1)}(a, a)$, and $A_{1,7}^{(1)}(a, a)$ are pairwise distinct for all $a \in [0 : 4)$.

Clearly,

$$\begin{aligned}
&\{A_{1,0}^{(0)}(a, a), A_{1,2}^{(0)}(a, a), A_{1,0}^{(1)}(a, a), A_{1,7}^{(1)}(a, a)\} \\
&= \begin{cases} \{y_0 x_0 \lambda_{0,0}, y_0 x_2 \lambda_{2,0}, y_1 x_0 \lambda_{0,0}, y_1 x_7 \lambda_{3,0}\}, & \text{if } a = 0, \\ \{y_0 x_0 \lambda_{0,0}, y_0 x_2 \lambda_{2,0}, y_1 x_0 \lambda_{0,0}, y_1 x_7 \lambda_{3,1}\}, & \text{if } a = 1, \\ \{y_0 x_0 \lambda_{0,1}, y_0 x_2 \lambda_{2,1}, y_1 x_0 \lambda_{0,1}, y_1 x_7 \lambda_{3,0}\}, & \text{if } a = 2, \\ \{y_0 x_0 \lambda_{0,1}, y_0 x_2 \lambda_{2,1}, y_1 x_0 \lambda_{0,1}, y_1 x_7 \lambda_{3,1}\}, & \text{if } a = 3. \end{cases}
\end{aligned}$$

- By (15), we have

$$\begin{aligned}
&\{y_0 x_j \lambda_{j,u} | j = 0, 2, u = 0, 1\} \\
&\cap \{y_1 x_j \lambda_{j \% 4,0} | j = 0, 7, u = 0, 1\} = \emptyset,
\end{aligned}$$

$$\begin{aligned}
& \left| (i_0 - i_1)(r + \delta - 1)(w + 2)/2 + \left(\left\lfloor \frac{l}{m} \right\rfloor - \left\lfloor \frac{j}{m} \right\rfloor \right) \frac{m}{2}(w + 2) + (l' - j')(w + 2) + u - u' + (g_0 - g_1)w \right| \\
& \leq (\mu - 1)(r + \delta - 1)(w + 2)/2 + \left(\frac{r + \delta - 1}{m} - 1 \right) \frac{m}{2}(w + 2) + \left(\frac{m}{2} - 1 \right)(w + 2) + w - 1 + w \\
& \leq \mu(r + \delta - 1)(w + 2)/2 + w - 3 \\
& \leq n(w + 2)/2 - 1 \\
& < q - 1.
\end{aligned} \tag{21}$$

which implies

$$\{A_{1,0}^{(0)}(a, a), A_{1,2}^{(0)}(a, a)\} \cap \{A_{1,0}^{(1)}(a, a), A_{1,7}^{(1)}(a, a)\} = \emptyset$$

for $a \in [0 : 4)$.

- According to i) of Lemma 5, we have $x_0\lambda_{0,u} \neq x_7\lambda_{3,u'}$ for $u, u' = 0, 1$, indicating $A_{1,0}^{(1)}(a, a) \neq A_{1,7}^{(1)}(a, a)$ for $a \in [0 : 4)$.
- Similarly, by ii) of Lemma 5, we have $A_{1,0}^{(0)}(a, a) \neq A_{1,2}^{(0)}(a, a)$ for $a \in [0 : 4)$.

The above analysis confirms that $A_{1,0}^{(0)}(a, a)$, $A_{1,2}^{(0)}(a, a)$, $A_{1,0}^{(1)}(a, a)$, and $A_{1,7}^{(1)}(a, a)$ are pairwise distinct for all $a \in [0 : 4)$.

Next, by Theorem 2, the $(n = 16, k = 10)$ vector LRC can be defined over the finite field $\mathbf{F}_{33}$. Let c be a primitive element over $\mathbf{F}_{33}$, then we can set

$$y_0 = 1, y_1 = c^{16},$$

and

$$x_i = \begin{cases} 1, & \text{if } i \in [0 : 4), \\ c^8, & \text{if } i \in [4 : 8). \end{cases}$$

Additionally,

$$\lambda_{0,u} = c^u, \lambda_{1,u} = c^{4+u}, \lambda_{2,u} = c^{2+u}, \lambda_{3,u} = c^{6+u}, u = 0, 1.$$

IV. A NEW CONSTRUCTION OF VECTOR LRCs WITH SMALL FILED SIZE AND SUB-PACKETIZATION LEVEL

In this section, we present the second construction of vector LRCs with a small sub-packetization level and repair bandwidth, which requires a smaller field size compared with Construction 1.

Construction 2. For any $r, \delta > 2$, $\mu > 1$, $s \geq 1$, $1 \leq m \leq r + \delta - 1$, and $1 < w \leq \delta - 1$, define an $(n = \mu(r + \delta - 1), k = \mu r - s)$ vector LRC over $\mathbf{F}_q$ with the parity-check matrix having the form as in (7)–(9) and sub-packetization level $\ell = w^m$, where $A_{t,j}^{(i)}$ in (8) and (9) satisfies

$$A_{t,j}^{(i)} = (A_j^{(i)})^t \text{ and } \begin{pmatrix} V_{j,0} \\ V_{j,1} \\ \vdots \\ V_{j,w-1} \end{pmatrix} A_j^{(i)} = \begin{pmatrix} \lambda_{j,0}^{(i)} V_{j,0} \\ \lambda_{j,1}^{(i)} V_{j,1} \\ \vdots \\ \lambda_{j,w-1}^{(i)} V_{j,w-1} \end{pmatrix} \tag{22}$$

with $\lambda_{j,t}^{(i)} \in \mathbf{F}_q$, and $V_{j,0}, V_{j,1}, \dots, V_{j,w-1}$ being defined by (11) and (12). For convenience, we assume that $m \mid r + \delta - 1$. Note that the parameter m controls the tradeoff between the sub-packetization level and the repair bandwidth.

In Construction 2, we employ the MDS array code $\mathcal{C}_5$ in [44] when $w = \delta - 1$ and $\mathcal{C}'_2$ in [45] when $w < \delta - 1$ as the local codes, i.e., H_i in (7) is exactly the parity-check matrix of either the MDS array code $\mathcal{C}_5$ in [44] or $\mathcal{C}'_2$ in [45]. For the global parity checks in (9), we elaborately design each block in (9) as powers of the corresponding blocks in H_i so that the matrix by vertically concatenating H_i and P_i forms a transpose of block Vandermonde matrix, this greatly facilitates the verification of the minimum distance using Lemma 2.

Lemma 6. The repair bandwidth of the vector LRC in Construction 2 is $(1 + \frac{(r+\delta-1-1)(w-1)}{d})\gamma^*$ if $\lambda_{j,0}^{(i)}, \lambda_{j,1}^{(i)}, \dots, \lambda_{j,w-1}^{(i)}$ are pairwise distinct for each $j \in [0 : r + \delta - 1)$, where $d = r + w - 1$, $\gamma^* = \frac{\ell}{w}d$ is the minimum repair bandwidth of an $(r + \delta - 1, r)$ MSR code with repair degree d and sub-packetization ℓ .

Proof. Note that nodes in local group i are subject to the parity-check matrix H_i , which is also the parity-check matrix of the MDS array code $\mathcal{C}_5$ in [44] when $w = \delta - 1$ and $\mathcal{C}'_2$ in [45] when $w < \delta - 1$, thus from [44, Theorem 13] and [45, Theorem 7], we have the desired result. $\square$

Theorem 3. The vector LRC in Construction 2 has the (r, δ) locality and is optimal w.r.t. the bound in (6) if $s < r$ and

- $\lambda_{j_0,u}^{(i_0)} \neq \lambda_{j_1,v}^{(i_1)}$ for all $u, v \in [0 : w)$ and $i_0, i_1 \in [0 : \mu)$, $j_0, j_1 \in [0 : r + \delta - 1)$ with $j_0 \not\equiv j_1 \pmod{m}$,
- $\lambda_{j_0,u}^{(i_0)} \neq \lambda_{j_1,u}^{(i_1)}$ for all $u \in [0 : w)$, $i_0, i_1 \in [0 : \mu)$, $j_0, j_1 \in [0 : r + \delta - 1)$ with $j_0 \equiv j_1 \pmod{m}$.

Proof. Note that if (i) and (ii) are satisfied, then $A_{j_0}^{(i_0)} A_{j_1}^{(i_1)} = A_{j_1}^{(i_1)} A_{j_0}^{(i_0)}$ and $A_{j_0}^{(i_0)} - A_{j_1}^{(i_1)}$ is non-singular for $i_0, i_1 \in [0 : \mu)$ and $j_0, j_1 \in [0 : r + \delta - 1)$ with $(i_0, j_0) \neq (i_1, j_1)$. The proof is the same as that of [44, Theorem 14]; therefore, we omit it here. In conjunction with Lemmas 1 and 2, we have that each local code of the vector LRC in Construction 2 is an $(r + \delta - 1, r)$ MDS array code, and thus the vector LRC in Construction 2 has the (r, δ) locality.

Thus, we only need to verify the requirement in Lemma 4, which is similar to the proof of Theorem 1 by applying Lemma 2. Therefore, we omit it here. $\square$

Theorem 4. The vector LRC generated by Construction 2 is an optimal LRC with (r, δ) locality and repair bandwidth $(1 + \frac{(r+\delta-1-1)(w-1)}{d})\gamma^*$ provided that q is a prime power satisfying (23), where $d = r + w - 1$, $\gamma^* = \frac{\ell}{w}d$ is the minimum repair bandwidth of a $(r + \delta - 1, r)$ MSR code with repair degree d and sub-packetization level ℓ .

$$q > \begin{cases} \mu(wm(\lceil \frac{r+\delta-1}{wm} \rceil - 1) + ((r+\delta-1)\%m)w), & \text{if } 0 < (r+\delta-1)\%(wm) < m, \\ \mu wm \lceil \frac{r+\delta-1}{wm} \rceil, & \text{otherwise.} \end{cases} \quad (23)$$

Proof. It suffices to verify the requirements in Lemma 6 and Theorem 3.

If $0 < (r+\delta-1)\%(wm) < m$, then

$$\lceil \frac{r+\delta-1}{wm} \rceil - 1 = \lfloor \frac{r+\delta-1}{wm} \rfloor$$

and

$$(r+\delta-1)\%(wm) = (r+\delta-1)\%m.$$

Let $\xi_{j',v}^{(i,z)}$, $i \in [0:\mu)$, $z \in [0:\lfloor \frac{r+\delta-1}{wm} \rfloor]$, $j' \in [0:m)$, $v \in [0:w)$, and $\xi_{j',v}^{(i,\lfloor \frac{r+\delta-1}{wm} \rfloor)}$, $i \in [0:\mu)$, $j' \in [0:(r+\delta-1)\%m)$, $v \in [0:w)$ be $\mu(wm\lfloor \frac{r+\delta-1}{wm} \rfloor + ((r+\delta-1)\%m)w)$ pairwise distinct nonzero elements in $\mathbf{F}_q$; Otherwise, let $\xi_{j',v}^{(i,z)}$, $i \in [0:\mu)$, $z \in [0:\lceil \frac{r+\delta-1}{wm} \rceil]$, $j' \in [0:m)$, $v \in [0:w)$ be $\mu wm \lceil \frac{r+\delta-1}{wm} \rceil$ pairwise distinct nonzero elements in $\mathbf{F}_q$. Now, for $j = zwm + um + j'$, $j' \in [0:m)$, $u \in [0:w)$, $z \in [0:\lceil \frac{r+\delta-1}{wm} \rceil]$, if we set

$$\lambda_{j,t}^{(i)} = \xi_{j',t \oplus u}^{(i,z)}$$

for $i \in [0:\mu)$, $j \in [0:r+\delta-1)$ and $t \in [0:w)$, where the subscript $t \oplus u$ is computed modulo w , then, it is easy to verify that the requirements in Lemma 6 and Theorem 3 can be satisfied, i.e., the desirable result follows. $\square$

In the following, we give a toy example of the vector LRC in Construction 2.

Example 2. Let $w = \delta - 1 = \mu = 2$, $s = m = 3$, $r = 4$, and let $c = 2$ be a primitive element in $\mathbf{F}_{13}$, then the parity-check matrix of the $(n = 12, k = 5)$ vector LRC with sub-packetization level $\ell = 8$ in Construction 2 can be defined over $\mathbf{F}_{13}$ with the parity-check matrix being

$$\mathcal{H} = \begin{pmatrix} I & \cdots & I \\ A_0^{(0)} & \cdots & A_5^{(0)} \\ & \ddots & \\ & & I & \cdots & I \\ & & A_0^{(1)} & \cdots & A_5^{(1)} \\ (A_0^{(0)})^2 & \cdots & (A_5^{(0)})^2 & (A_0^{(1)})^2 & \cdots & (A_5^{(1)})^2 \\ (A_0^{(0)})^3 & \cdots & (A_5^{(0)})^3 & (A_0^{(1)})^3 & \cdots & (A_5^{(1)})^3 \\ (A_0^{(0)})^4 & \cdots & (A_5^{(0)})^4 & (A_0^{(1)})^4 & \cdots & (A_5^{(1)})^4 \end{pmatrix}. \quad (24)$$

Since $(r+\delta-1)\%(wm) = 0$ in this case, then according to the analysis in the proof of Theorem 4, let

$$\xi_{j',v}^{(i,0)} = c^{i+2j'+v}, j' \in [0:3), i, v \in [0:2)$$

be 12 pairwise distinct nonzero elements in $\mathbf{F}_{13}$. For $j = 3u + j'$, $j' \in [0:3)$, $u \in [0:2)$, we set

$$\lambda_{j,t}^{(i)} = \xi_{j',t \oplus u}^{(i,0)}$$

for $i, t \in [0:2)$ and $j \in [0:6)$. From which we obtain

$$\lambda_{j,0}^{(0)} = \lambda_{j+3,1}^{(0)} = c^{2j} \neq c^{2j+1} = \lambda_{j,1}^{(0)} = \lambda_{j+3,0}^{(0)} \text{ for } j = 0, 1, 2,$$

and

$$\lambda_{j,0}^{(1)} = \lambda_{j+3,1}^{(1)} = c^{2j+6} \neq c^{2j+7} = \lambda_{j,1}^{(1)} = \lambda_{j+3,0}^{(1)} \text{ for } j = 0, 1, 2,$$

i.e., the condition in Lemma 6 is satisfied. Furthermore, for $j_0, j_1 \in [0:6)$, $u, v \in [0:2)$ and $i_0, i_1 \in [0:2)$, we see that $\lambda_{j_0,u}^{(i_0)} \neq \lambda_{j_1,v}^{(i_1)}$ if $j_0 \not\equiv j_1 \pmod 3$ and $\lambda_{j_0,u}^{(i_0)} \neq \lambda_{j_1,u}^{(i_1)}$ otherwise, i.e., conditions i) and ii) of Theorem 3 are satisfied. Furthermore, the blocks $A_j^{(i)}$, $j \in [0:5)$ and $i \in [0:2)$ in (24) are

$$\begin{aligned} A_0^{(0)} &= \begin{pmatrix} e_0 \\ e_1 \\ e_2 \\ e_3 \\ ce_4 \\ ce_5 \\ ce_6 \\ ce_7 \end{pmatrix}, A_1^{(0)} = \begin{pmatrix} c^2 e_0 \\ c^2 e_1 \\ c^3 e_2 \\ c^3 e_3 \\ c^2 e_4 \\ c^2 e_5 \\ c^3 e_6 \\ c^3 e_7 \end{pmatrix}, A_2^{(0)} = \begin{pmatrix} c^4 e_0 \\ c^5 e_1 \\ c^4 e_2 \\ c^5 e_3 \\ c^4 e_4 \\ c^5 e_5 \\ c^4 e_6 \\ c^5 e_7 \end{pmatrix}, \\ A_3^{(0)} &= \begin{pmatrix} ce_0 \\ ce_1 \\ ce_2 \\ ce_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \end{pmatrix}, A_4^{(0)} = \begin{pmatrix} c^3 e_0 \\ c^3 e_1 \\ c^2 e_2 \\ c^2 e_3 \\ c^3 e_4 \\ c^3 e_5 \\ c^2 e_6 \\ c^2 e_7 \end{pmatrix}, A_5^{(0)} = \begin{pmatrix} c^5 e_0 \\ c^4 e_1 \\ c^5 e_2 \\ c^4 e_3 \\ c^5 e_4 \\ c^4 e_5 \\ c^5 e_6 \\ c^4 e_7 \end{pmatrix}, \\ A_0^{(1)} &= \begin{pmatrix} c^6 e_0 \\ c^6 e_1 \\ c^6 e_2 \\ c^6 e_3 \\ c^7 e_4 \\ c^7 e_5 \\ c^7 e_6 \\ c^7 e_7 \end{pmatrix}, A_1^{(1)} = \begin{pmatrix} c^8 e_0 \\ c^8 e_1 \\ c^9 e_2 \\ c^9 e_3 \\ c^8 e_4 \\ c^8 e_5 \\ c^9 e_6 \\ c^9 e_7 \end{pmatrix}, A_2^{(1)} = \begin{pmatrix} c^{10} e_0 \\ c^{11} e_1 \\ c^{10} e_2 \\ c^{11} e_3 \\ c^{10} e_4 \\ c^{11} e_5 \\ c^{10} e_6 \\ c^{11} e_7 \end{pmatrix}, \\ A_3^{(1)} &= \begin{pmatrix} c^7 e_0 \\ c^7 e_1 \\ c^7 e_2 \\ c^7 e_3 \\ c^6 e_4 \\ c^6 e_5 \\ c^6 e_6 \\ c^6 e_7 \end{pmatrix}, A_4^{(1)} = \begin{pmatrix} c^9 e_0 \\ c^9 e_1 \\ c^8 e_2 \\ c^8 e_3 \\ c^9 e_4 \\ c^9 e_5 \\ c^8 e_6 \\ c^8 e_7 \end{pmatrix}, A_5^{(1)} = \begin{pmatrix} c^{11} e_0 \\ c^{10} e_1 \\ c^{11} e_2 \\ c^{10} e_3 \\ c^{11} e_4 \\ c^{10} e_5 \\ c^{11} e_6 \\ c^{10} e_7 \end{pmatrix}. \end{aligned}$$

Next, we verify that the code is optimal w.r.t. the bound in (6), i.e., it can tolerate arbitrary 5 node failures. Let $\mathbf{f}_0^{(0)}, \dots, \mathbf{f}_5^{(0)}, \mathbf{f}_0^{(1)}, \dots, \mathbf{f}_5^{(1)}$ denote the data stored at the 12 nodes, respectively. Suppose nodes 0, 1, 6, 7, and 8 fail. From the parity-check matrix in (24), we have

$$\begin{pmatrix} I & I & \cdots & I \\ A_0^{(0)} & A_1^{(0)} & \cdots & A_5^{(0)} \end{pmatrix} \begin{pmatrix} \mathbf{f}_0^{(0)} \\ \mathbf{f}_1^{(0)} \\ \vdots \\ \mathbf{f}_5^{(0)} \end{pmatrix} = \mathbf{0}, \quad (25)$$

and

$$\begin{pmatrix} I & \dots & I \\ A_0^{(1)} & \dots & A_5^{(1)} \\ (A_0^{(0)})^2 & \dots & (A_5^{(0)})^2 \end{pmatrix} \begin{pmatrix} \mathbf{f}_0^{(0)} \\ \vdots \\ \mathbf{f}_5^{(0)} \\ \mathbf{f}_0^{(1)} \\ \vdots \\ \mathbf{f}_5^{(1)} \end{pmatrix} = \mathbf{0},$$

from which we can obtain

$$\begin{pmatrix} \mathbf{f}_0^{(0)} \\ \mathbf{f}_1^{(0)} \end{pmatrix} = - \begin{pmatrix} I & I \\ A_0^{(0)} & A_1^{(0)} \end{pmatrix}^{-1} \begin{pmatrix} I & I & I & I \\ A_2^{(0)} & A_3^{(0)} & A_4^{(0)} & A_5^{(0)} \end{pmatrix} \begin{pmatrix} \mathbf{f}_2^{(0)} \\ \mathbf{f}_3^{(0)} \\ \mathbf{f}_4^{(0)} \\ \mathbf{f}_5^{(0)} \end{pmatrix},$$

and (26) on the next page. Since it can be verified that the matrices

$$\begin{pmatrix} I & I \\ A_0^{(0)} & A_1^{(0)} \end{pmatrix} \text{ and } \begin{pmatrix} I & I & I \\ A_0^{(1)} & A_1^{(1)} & A_2^{(1)} \\ (A_0^{(1)})^2 & (A_1^{(1)})^2 & (A_2^{(1)})^2 \end{pmatrix}$$

are invertible by Lemma 2. Therefore, the data in nodes 0, 1, 6, 7, and 8 can be reconstructed.

Finally, we illustrate the node repair process. W.L.O.G., consider the repair of node 0, from (24), we know that the data $\mathbf{f}_0^{(0)}$ in node 0 subject to (25). By multiplying

$$\begin{pmatrix} e_0 + e_4 \\ e_1 + e_5 \\ e_2 + e_6 \\ e_3 + e_7 \end{pmatrix}$$

from the left to both sides of (25), we obtain the following equations

$$\begin{pmatrix} e_0 + e_4 \\ e_1 + e_5 \\ e_2 + e_6 \\ e_3 + e_7 \\ e_0 + ce_4 \\ e_1 + ce_5 \\ e_2 + ce_6 \\ e_3 + ce_7 \end{pmatrix} \mathbf{f}_0^{(0)} = - \begin{pmatrix} e_0 + e_4 \\ e_1 + e_5 \\ e_2 + e_6 \\ e_3 + e_7 \\ ce_0 + e_4 \\ ce_1 + e_5 \\ ce_2 + e_6 \\ ce_3 + e_7 \end{pmatrix} \mathbf{f}_3^{(0)} - \sum_{j=1, j \neq 3}^5 \Lambda_j \begin{pmatrix} e_0 + e_4 \\ e_1 + e_5 \\ e_2 + e_6 \\ e_3 + e_7 \\ e_0 + e_4 \\ e_1 + e_5 \\ e_2 + e_6 \\ e_3 + e_7 \end{pmatrix} \mathbf{f}_j^{(0)},$$

where Λ_j represents a known diagonal matrix over $\mathbf{F}_{13}$ for $j \in [1, 6] \setminus \{3\}$. As an example, $\Lambda_1 = \text{diag}(1, 1, 1, 1, c^2, c^2, c^3, c^3)$.

Based on the above equations, we can solve $\mathbf{f}_0^{(0)}$ since the coefficient matrix has rank 8, the right-hand side can be

eliminated by connecting the $d = 5$ remaining nodes in local group 0, and downloading $\mathbf{f}_3^{(0)}$, $(e_0 + e_4)\mathbf{f}_j^{(0)}$, $(e_1 + e_5)\mathbf{f}_j^{(0)}$, $(e_2 + e_6)\mathbf{f}_j^{(0)}$, $(e_3 + e_7)\mathbf{f}_j^{(0)}$ for $j \in [1, 6] \setminus \{3\}$, where the repair bandwidth is $8 + 4 \times 4 = 24$, which is consistent with the result obtained in Lemma 6.

V. A NEW CONSTRUCTION OF VECTOR LRCs WITH SUB-PACKETIZATION LEVEL 2

This section proposes a vector LRC with sub-packetization level 2.

Construction 3. For convenience of notation, we assume that $r + \delta - 1$ is even and $r > \delta$, where $\delta \geq 3$. Construct an $(n = \mu(r + \delta - 1), k = \mu r - s)$ vector LRC over $\mathbf{F}_q$ with the parity-check matrix having the form as in (7)–(9) and sub-packetization level $\ell = 2$, where $A_{t,j}^{(i)}$ in (8) and (9) is defined as

$$A_{t,j}^{(i)} = \begin{cases} \begin{pmatrix} \lambda_{i,j}^t & 1 \\ 0 & \lambda_{i,j}^t \end{pmatrix}, & \text{if } t \in [0 : \delta - 1], 2 \mid j, \\ \begin{pmatrix} \lambda_{i,j}^t & 0 \\ 0 & \lambda_{i,j}^t \end{pmatrix}, & \text{if } t \in [0 : \delta - 1], 2 \nmid j, \\ \lambda_{i,j}^t I, & \text{if } t \in [\delta - 1 : \delta + s - 1], \end{cases} \quad (27)$$

and $\lambda_{i,j}, i \in [0 : \mu], j \in [0 : r + \delta - 1]$ are pairwise distinct elements in $\mathbf{F}_q \setminus \{0, 1\}$.

In Construction 3, the parity-check matrix H_i of the local code defined by (8) and (27) is, in fact, the parity-check matrix of an $(r + \delta - 1, r)$ MDS array code with sub-packetization level 2, which subsumes the one in [41, Construction 3] as a special case, where the latter only works for $\delta = 3$. For the global parity checks in (9), similar to Construction 1, we elaborately design each block in (9) as diagonal matrices.

Theorem 5. The repair bandwidth of the vector LRC in Construction 3 is $\frac{3r + \delta - 1}{2}$.

Proof. W.L.O.G., we consider the repair bandwidth of nodes in the local group 0 since the other cases are similar.

Let $\mathbf{f}_0, \dots, \mathbf{f}_{r+\delta-2}$ denote the data stored in the $r + \delta - 1$ nodes of local group 0. Then local code 0 is subject to the following parity-check equations according to (7) and (8)

$$A_{0,0}^{(0)} \mathbf{f}_0 + A_{0,1}^{(0)} \mathbf{f}_1 + \dots + A_{0,r+\delta-2}^{(0)} \mathbf{f}_{r+\delta-2} = \mathbf{0}, \quad (28)$$

$$A_{1,0}^{(0)} \mathbf{f}_0 + A_{1,1}^{(0)} \mathbf{f}_1 + \dots + A_{1,r+\delta-2}^{(0)} \mathbf{f}_{r+\delta-2} = \mathbf{0}, \quad (29)$$

$\vdots$

$$A_{\delta-2,0}^{(0)} \mathbf{f}_0 + A_{\delta-2,1}^{(0)} \mathbf{f}_1 + \dots + A_{\delta-2,r+\delta-2}^{(0)} \mathbf{f}_{r+\delta-2} = \mathbf{0}. \quad (30)$$

Suppose that node i ($i \in [0 : r + \delta - 1]$) fails. We can regenerate it by contacting some $d = r + 1$ surviving nodes in the local group 0, where nodes with indices in the set $T = \{j | j \equiv i \pmod{2}, j \neq i, 0 \leq j < r + \delta - 1\}$ are compulsory while the other $d - |T|$ nodes can be arbitrarily chosen. Let L denote the indices of these $d - |T|$ nodes, i.e., L is an arbitrary subset of $\{j | j \not\equiv i \pmod{2}, 0 \leq j < r + \delta - 1\}$ with $|L| = d - |T| = \frac{r - \delta + 5}{2}$. Clearly, some $\delta - 3$ nodes are not connected, say nodes $u_0, u_1, \dots, u_{\delta-4}$ are not connected, where

$$0 \leq u_0 < u_1 < \dots < u_{\delta-4} < r + \delta - 1 \text{ and } u_j \not\equiv i \pmod{2}.$$

$$\begin{pmatrix} \mathbf{f}_0^{(1)} \\ \mathbf{f}_1^{(1)} \\ \mathbf{f}_2^{(1)} \end{pmatrix} = - \begin{pmatrix} I & I & I \\ A_0^{(1)} & A_1^{(1)} & A_2^{(1)} \\ (A_0^{(1)})^2 & (A_1^{(1)})^2 & (A_2^{(1)})^2 \end{pmatrix}^{-1} \left(\begin{pmatrix} I & I & I \\ A_3^{(1)} & A_4^{(1)} & A_5^{(1)} \\ (A_3^{(1)})^2 & (A_4^{(1)})^2 & (A_5^{(1)})^2 \end{pmatrix} \begin{pmatrix} \mathbf{f}_3^{(1)} \\ \mathbf{f}_4^{(1)} \\ \mathbf{f}_5^{(1)} \end{pmatrix} - \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \\ \sum_{j=0}^5 (A_j^{(0)})^2 \mathbf{f}_j^{(0)} \end{pmatrix} \right). \quad (26)$$

In the following, we prove that node i can be regenerated by downloading

$$\mathbf{f}_j, j \in T, e_0 \mathbf{f}_l, l \in L,$$

where $e_0 = \begin{pmatrix} 1 & 0 \end{pmatrix}$.

- If $2 \mid i$, by multiplying e_0 with (28), (29), and (30) from the left, we obtain

$$\begin{pmatrix} e_0 A_{0,i}^{(0)} \\ e_0 A_{1,i}^{(0)} \\ \vdots \\ e_0 A_{\delta-2,i}^{(0)} \end{pmatrix} \mathbf{f}_i + \sum_{j \in T} \begin{pmatrix} e_0 A_{0,j}^{(0)} \\ e_0 A_{1,j}^{(0)} \\ \vdots \\ e_0 A_{\delta-2,j}^{(0)} \end{pmatrix} \mathbf{f}_j + \sum_{j=1,2 \nmid j}^{r+\delta-2} \begin{pmatrix} e_0 A_{0,j}^{(0)} \\ e_0 A_{1,j}^{(0)} \\ \vdots \\ e_0 A_{\delta-2,j}^{(0)} \end{pmatrix} \mathbf{f}_j = \mathbf{0}. \quad (31)$$

For $t \in [0 : \delta - 1]$, by (27), we have

$$e_0 A_{t,j}^{(0)} = \begin{cases} (\lambda_{0,j}^t & 1), & \text{if } 2 \mid j, \\ (\lambda_{0,j}^t & 0), & \text{otherwise,} \end{cases} \quad (32)$$

which together with (31) leads to

$$\begin{pmatrix} 1 & 1 \\ \lambda_{0,i} & 1 \\ \vdots & \vdots \\ \lambda_{0,i}^{\delta-2} & 1 \end{pmatrix} \mathbf{f}_i + \sum_{j \in T} \begin{pmatrix} 1 & 1 \\ \lambda_{0,j} & 1 \\ \vdots & \vdots \\ \lambda_{0,j}^{\delta-2} & 1 \end{pmatrix} \mathbf{f}_j + \sum_{j \in L} \begin{pmatrix} 1 & 0 \\ \lambda_{0,j} & 0 \\ \vdots & \vdots \\ \lambda_{0,j}^{\delta-2} & 0 \end{pmatrix} \mathbf{f}_j + \sum_{j=0}^{\delta-4} \begin{pmatrix} 1 & 0 \\ \lambda_{0,u_j} & 0 \\ \vdots & \vdots \\ \lambda_{0,u_j}^{\delta-2} & 0 \end{pmatrix} \mathbf{f}_{u_j} = \mathbf{0},$$

which is equivalent to

$$\begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ \lambda_{0,i} & 1 & \lambda_{0,u_0} & \cdots & \lambda_{0,u_{\delta-4}} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \lambda_{0,i}^{\delta-2} & 1 & \lambda_{0,u_0}^{\delta-2} & \cdots & \lambda_{0,u_{\delta-4}}^{\delta-2} \end{pmatrix} \begin{pmatrix} \mathbf{f}_i \\ e_0 \mathbf{f}_{u_0} \\ \vdots \\ e_0 \mathbf{f}_{u_{\delta-4}} \end{pmatrix} = - \sum_{j \in T} \begin{pmatrix} 1 & 1 \\ \lambda_{0,j} & 1 \\ \vdots & \vdots \\ \lambda_{0,j}^{\delta-2} & 1 \end{pmatrix} \mathbf{f}_j - \sum_{j \in L} \begin{pmatrix} 1 \\ \lambda_{0,j} \\ \vdots \\ \lambda_{0,j}^{\delta-2} \end{pmatrix} e_0 \mathbf{f}_j,$$

from which $\mathbf{f}_i$ can be obtained because the RHS is known from the download data and the coefficient matrix is non-singular since $1, \lambda_{0,i}, \lambda_{0,u_0}, \dots, \lambda_{0,u_{\delta-4}}$ are pairwise distinct.

Thus the repair bandwidth γ_i of node i is

$$\begin{aligned} \gamma_i &= 2|T| + L \\ &= 2\left(\frac{r+\delta-1}{2} - 1\right) + \frac{r-\delta+5}{2} \\ &= \frac{3r+\delta-1}{2}. \end{aligned}$$

- If $2 \nmid i$, then let $e_1 = \begin{pmatrix} 0 & 1 \end{pmatrix}$. By $e_0 \times (28) - e_1 \times (28)$, $e_0 \times (29) - e_1 \times (28)$, ..., $e_0 \times (30) - e_1 \times (28)$, we obtain

$$\begin{pmatrix} e_0 A_{0,i}^{(0)} - e_1 A_{0,i}^{(0)} \\ e_0 A_{1,i}^{(0)} - e_1 A_{0,i}^{(0)} \\ \vdots \\ e_0 A_{\delta-2,i}^{(0)} - e_1 A_{0,i}^{(0)} \end{pmatrix} \mathbf{f}_i + \sum_{j \in T} \begin{pmatrix} e_0 A_{0,j}^{(0)} - e_1 A_{0,j}^{(0)} \\ e_0 A_{1,j}^{(0)} - e_1 A_{0,j}^{(0)} \\ \vdots \\ e_0 A_{\delta-2,j}^{(0)} - e_1 A_{0,j}^{(0)} \end{pmatrix} \mathbf{f}_j + \sum_{j=0,2 \nmid j}^{r+\delta-2} \begin{pmatrix} e_0 A_{0,j}^{(0)} - e_1 A_{0,j}^{(0)} \\ e_0 A_{1,j}^{(0)} - e_1 A_{0,j}^{(0)} \\ \vdots \\ e_0 A_{\delta-2,j}^{(0)} - e_1 A_{0,j}^{(0)} \end{pmatrix} \mathbf{f}_j = \mathbf{0}. \quad (33)$$

By (27), we have $e_1 A_{0,j}^{(0)} = \begin{pmatrix} 0 & 1 \end{pmatrix}$, which together with (32) and (33) gives

$$\begin{pmatrix} 1 & -1 \\ \lambda_{0,i} & -1 \\ \vdots & \vdots \\ \lambda_{0,i}^{\delta-2} & -1 \end{pmatrix} \mathbf{f}_i + \sum_{j \in T} \begin{pmatrix} 1 & -1 \\ \lambda_{0,j} & -1 \\ \vdots & \vdots \\ \lambda_{0,j}^{\delta-2} & -1 \end{pmatrix} \mathbf{f}_j + \sum_{j \in L} \begin{pmatrix} 1 & 0 \\ \lambda_{0,j} & 0 \\ \vdots & \vdots \\ \lambda_{0,j}^{\delta-2} & 0 \end{pmatrix} \mathbf{f}_j + \sum_{j=0}^{\delta-4} \begin{pmatrix} 1 & 0 \\ \lambda_{0,u_j} & 0 \\ \vdots & \vdots \\ \lambda_{0,u_j}^{\delta-2} & 0 \end{pmatrix} \mathbf{f}_{u_j} = \mathbf{0}.$$

Similar to the case of $2 \mid i$, $\mathbf{f}_i$ can be solved from the above equations with the downloaded data and the repair bandwidth γ_i of node i is $\gamma_i = \frac{3r+\delta-1}{2}$.

This completes the proof. $\square$

Theorem 6. The vector LRC in Construction 3 has the (r, δ) locality and is optimal w.r.t. the bound in (6) if $s < r$.

Proof. By Lemma 4, it suffices to show the requirement in Lemma 4 is satisfied and each local code of the vector LRC in Construction 3 is an $(r + \delta - 1, r)$ MDS array code.

i) We first verify the requirement in Lemma 4, i.e.,

$$\text{rank} \left(\begin{pmatrix} H_{i_0}[T_0] & H_{i_1}[T_1] & \cdots & H_{i_{b-1}}[T_{b-1}] \\ P_{i_0}[T_0] & P_{i_1}[T_1] & \cdots & P_{i_{b-1}}[T_{b-1}] \end{pmatrix} \right) = 2(\delta + s - 1)$$

for any $0 \leq i_0 < i_1 < \cdots < i_{b-1} < \mu$, $T_j \subset [0 : r + \delta - 1]$ and $|T_j| > \delta - 1$ for $0 \leq j < b$ such that $|T_0| + |T_1| + \cdots + |T_{b-1}| = \delta + s - 1$. In the following, we verify the case of $b = 1$.

For any $i \in [0 : \mu)$ and $J \subset [0 : r + \delta - 1)$ with $|J| = \delta + s - 1$, say $J = \{j_0, \dots, j_{\delta+s-2}\}$, we have

$$\begin{pmatrix} H_i[J] \\ P_i[J] \end{pmatrix} = \begin{pmatrix} A_{0,j_0}^{(i)} & A_{0,j_1}^{(i)} & \dots & A_{0,j_{\delta+s-2}}^{(i)} \\ A_{1,j_0}^{(i)} & A_{1,j_1}^{(i)} & \dots & A_{1,j_{\delta+s-2}}^{(i)} \\ \vdots & \vdots & \ddots & \vdots \\ A_{\delta-2,j_0}^{(i)} & A_{\delta-2,j_1}^{(i)} & \dots & A_{\delta-2,j_{\delta+s-2}}^{(i)} \\ \lambda_{i,j_0}^{\delta-1} I & \lambda_{i,j_1}^{\delta-1} I & \dots & \lambda_{i,j_{\delta+s-2}}^{\delta-1} I \\ \lambda_{i,j_0}^{\delta} I & \lambda_{i,j_1}^{\delta} I & \dots & \lambda_{i,j_{\delta+s-2}}^{\delta} I \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{i,j_0}^{\delta+s-2} I & \lambda_{i,j_1}^{\delta+s-2} I & \dots & \lambda_{i,j_{\delta+s-2}}^{\delta+s-2} I \end{pmatrix} = \begin{pmatrix} 1 & x_{0,0} & 1 & x_{0,1} & \dots & 1 & x_{0,\delta+s-2} \\ 0 & 1 & 0 & 1 & \dots & 0 & 1 \\ \lambda_{i,j_0} & x_{1,0} & \lambda_{i,j_1} & x_{1,1} & \dots & \lambda_{i,j_{\delta+s-2}} & x_{1,\delta+s-2} \\ 0 & \lambda_{i,j_0} & 0 & \lambda_{i,j_1} & \dots & 0 & \lambda_{i,j_{\delta+s-2}} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \lambda_{i,j_0}^{\delta-2} & x_{\delta-2,0} & \lambda_{i,j_1}^{\delta-2} & x_{\delta-2,1} & \dots & \lambda_{i,j_{\delta+s-2}}^{\delta-2} & x_{\delta-2,\delta+s-2} \\ 0 & \lambda_{i,j_0}^{\delta-2} & 0 & \lambda_{i,j_1}^{\delta-2} & \dots & 0 & \lambda_{i,j_{\delta+s-2}}^{\delta-2} \\ \lambda_{i,j_0}^{\delta-1} & 0 & \lambda_{i,j_1}^{\delta-1} & 0 & \dots & \lambda_{i,j_{\delta+s-2}}^{\delta-1} & 0 \\ 0 & \lambda_{i,j_0}^{\delta-1} & 0 & \lambda_{i,j_1}^{\delta-1} & \dots & 0 & \lambda_{i,j_{\delta+s-2}}^{\delta-1} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \lambda_{i,j_0}^{\delta+s-2} & 0 & \lambda_{i,j_1}^{\delta+s-2} & 0 & \dots & \lambda_{i,j_{\delta+s-2}}^{\delta+s-2} & 0 \\ 0 & \lambda_{i,j_0}^{\delta+s-2} & 0 & \lambda_{i,j_1}^{\delta+s-2} & \dots & 0 & \lambda_{i,j_{\delta+s-2}}^{\delta+s-2} \end{pmatrix}$$

where $x_{t,j} = 0, 1$ for $0 \leq t < \delta - 1$ and $0 \leq j < \delta + s - 1$. By swapping the rows and columns, it is easy to see that

$\begin{pmatrix} H_i[J] \\ P_i[J] \end{pmatrix}$ is equivalent to the following matrix

$$B = \begin{pmatrix} 1 & 1 & \dots & 1 & x_{0,0} & x_{0,1} & \dots & x_{0,\delta+s-2} \\ \lambda_{i,j_0} & \lambda_{i,j_1} & \dots & \lambda_{i,j_{\delta+s-2}} & x_{1,0} & x_{1,1} & \dots & x_{1,\delta+s-2} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda_{i,j_0}^{\delta-2} & \lambda_{i,j_1}^{\delta-2} & \dots & \lambda_{i,j_{\delta+s-2}}^{\delta-2} & x_{\delta-2,0} & x_{\delta-2,1} & \dots & x_{\delta-2,\delta+s-2} \\ \lambda_{i,j_0}^{\delta-1} & \lambda_{i,j_1}^{\delta-1} & \dots & \lambda_{i,j_{\delta+s-2}}^{\delta-1} & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda_{i,j_0}^{\delta+s-2} & \lambda_{i,j_1}^{\delta+s-2} & \dots & \lambda_{i,j_{\delta+s-2}}^{\delta+s-2} & 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 1 & 1 & \dots & 1 \\ 0 & 0 & \dots & 0 & \lambda_{i,j_0} & \lambda_{i,j_1} & \dots & \lambda_{i,j_{\delta+s-2}} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \lambda_{i,j_0}^{\delta+s-2} & \lambda_{i,j_1}^{\delta+s-2} & \dots & \lambda_{i,j_{\delta+s-2}}^{\delta+s-2} \\ 0 & 0 & \dots & 0 & \lambda_{i,j_0}^{\delta+s-2} & \lambda_{i,j_1}^{\delta+s-2} & \dots & \lambda_{i,j_{\delta+s-2}}^{\delta+s-2} \end{pmatrix}$$

under elementary transformation. Note that B is a block upper triangular matrix, then we have

$$\begin{aligned} & \det(B) \\ &= \left(\det \begin{pmatrix} 1 & 1 & \dots & 1 \\ \lambda_{i,j_0} & \lambda_{i,j_1} & \dots & \lambda_{i,j_{\delta+s-2}} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{i,j_0}^{\delta+s-2} & \lambda_{i,j_1}^{\delta+s-2} & \dots & \lambda_{i,j_{\delta+s-2}}^{\delta+s-2} \end{pmatrix} \right)^2 \\ & \neq 0, \end{aligned}$$

where the inequality holds since $\lambda_{i,j_0}, \lambda_{i,j_1}, \dots, \lambda_{i,j_{\delta+s-2}}$ are pairwise distinct nonzero elements, we conclude that the matrix $\begin{pmatrix} H_i[J] \\ P_i[J] \end{pmatrix}$ is non-singular.

The case of $b > 1$ can be proved similarly by the fact $\lambda_{i,j}, i = 0, 1, \dots, \mu - 1, j = 0, 1, \dots, r + \delta - 2$ are pairwise distinct. Therefore, we omit it here.

ii) To prove that each local code of the vector LRC is an $(r + \delta - 1, r)$ MDS array code, it is equivalent to prove that $H_i[J]$ is non-singular for any $i \in [0 : \mu)$ and $J \subset [0 : r + \delta - 1)$ with $|J| = \delta - 1$ according to Lemma 1, which can be guaranteed if $\lambda_{i,j}, j = 0, 1, \dots, r + \delta - 2$ are pairwise distinct, the analysis is similar to i), therefore, we omit it here. $\square$

VI. COMPARISONS

In this section, we compare the vector LRCs proposed in this paper with some notable explicit ones in the literature, such as Constructions 5.1, 5.3 in [27], and Construction 1 in [28]. Note that these known constructions rely on the existence of an $(r + \delta - 1, r)$ MSR code with a flexible repair degree $d < r + \delta - 2$, which is scarce in the literature, e.g., the low code-rate MSR codes in [6] and the high code-rate MSR codes in [15], [21], [24]. We limit to high code-rate MSR codes as they can reduce the storage overhead. For convenience, we refer to the two MSR codes in [15] and the ones in [21], [24] as YB code 1, YB code 2, CB code, and LLT code, respectively. Table III gives the details.

Note that all the vector LRCs listed in Table III have the same minimum distance $d_{\min} = \delta + s$, which is optimal w.r.t. the bound in (6). It has been observed that the newly proposed vector LRCs have the following advantages:

- i) The newly proposed vector LRCs work for any number of local parities $\delta - 1$ and any number of global parities s as long as $s < r$, which provides some flexibility in contrast to the one in Construction 5.1 in [27] and the one in [38], as they are distance-optimal only when $s < \delta - 1$. A consequence of the restriction $s < \delta - 1$ is that when one wishes to increase the minimal distance $d_{\min} = s + \delta - 1$, δ and s may need to be increased. Note that the code-rate $\frac{k}{n} = \frac{\mu r}{\mu(r + \delta - 1) + s}$ or $\frac{\mu r - s}{\mu(r + \delta - 1)}$ drops more rapidly when increasing δ than increasing s .
- ii) The newly proposed vector LRCs have an all-symbol locality. In contrast, the explicit vector LRC in Construction 5.3 in [27] only has information locality, i.e., only a subset of code symbols have locality.
- iii) The proposed new vector LRCs have a small sub-packetization level, i.e., $w^{m/2}, w^m$ ($1 \leq m \leq r + \delta - 1$), and 2 for the new vector LRCs in Constructions 1–3, respectively, which are much smaller than existing vector LRCs.
- iv) The three new vector LRCs are built on small finite fields such that the required field sizes are around the code lengths, which are also much smaller than that of Construction 1 in [28].
- v) The new vector LRCs in Constructions 1 and 2 support any repair degree d in $[r + 1 : r + \delta - 1)$, whereas the one in [38] only supports $d = r + \delta - 2$.

TABLE III

COMPARISON OF KEY PARAMETERS AMONG NEW DISTANCE OPTIMAL (n, k) VECTOR LRCs AND EXISTING ONES IN [27], [28], [38], WHERE EACH LOCAL CODE IS A $(r + \delta - 1, r)$ MDS ARRAY CODE WITH REPAIR DEGREE $d = r + w - 1$ AND SUB-PACKETIZATION LEVEL ℓ , $1 < w \leq \delta - 1$,

$$k = n - \mu(\delta - 1) - s, \text{ AND } \epsilon = \frac{(\frac{r+\delta-1}{m}-1)(w-1)}{d}.$$

	Code Rate $\frac{k}{n}$	Sub-packetization level ℓ	Field size q	Repair bandwidth γ	Remarks
Construction 5.1 in [27] based on YB code 1 in [15] based on YB code 2 in [15] based on CB code in [21] based on LLT code in [24]	$\frac{\mu r}{\mu(r+\delta-1)+s}$	$w^{r+\delta+s-1}$ $w^{r+\delta+s-1}$ $w^{r+\delta+s-1}$ $w^{\lceil \frac{r+\delta+s-1}{2} \rceil}$	$\geq w(r+\delta+s-1)$ $\geq r+\delta+s$ $\geq r+\delta+s+w-2$ $\geq r+\delta+s+w^{\lceil \frac{r+\delta+s-1}{2} \rceil} - 1$	$d \frac{\ell}{w}$	$s < \delta - 1$
Construction 5.3 in [27] based on YB code 1 in [15] based on YB code 2 in [15] based on CB code in [21] based on LLT code in [24]	$\frac{\mu r}{\mu(r+\delta-1)+s}$	$w^{\mu r+\delta+s-1}$ $w^{\mu r+\delta+s-1}$ $w^{\mu r+\delta+s-1}$ $w^{\lceil \frac{\mu r+\delta+s-1}{2} \rceil}$	$\geq w(\mu r+\delta+s-1)$ $\geq \mu r+\delta+s$ $\geq \mu r+\delta+w+s-2$ $\geq \mu r+\delta+s-1$ $+w^{\lceil \frac{\mu r+\delta+s-1}{2} \rceil}$	$d \frac{\ell}{w}$	Inf. locality
Construction 1 in [28] based on YB code 1 in [15] based on YB code 2 in [15] based on CB code in [21] based on LLT code in [24]	$\frac{\mu r-s}{\mu(r+\delta-1)}$	$w^{r+\delta-1}$ $w^{r+\delta-1}$ $w^{r+\delta-1}$ $w^{\lceil \frac{r+\delta-1}{2} \rceil}$	$\geq (w(r+\delta-1))^{\mu r \ell}$ $\geq (r+\delta)^{\mu r \ell}$ $\geq (r+\delta+w-2)^{\mu r \ell}$ $\geq (r+\delta+w^{\lceil \frac{r+\delta-1}{2} \rceil} - 1)^{\mu r \ell}$	$d \frac{\ell}{w}$	
Codes with MSR locality [38]	$\frac{\mu r-s}{\mu(r+\delta-1)}$	$w^{\frac{r+\delta-1}{w}}$	$\geq \mu(r+\delta-1)$	$d \frac{\ell}{w}$	$s < w = \delta - 1, \delta - 1 \mid r$
New vector LRC in Construction 1	$\frac{\mu r-s}{\mu(r+\delta-1)}$	$w^{\frac{m}{2}}$	$> \begin{cases} n(w+2)/2, & w=2, \\ n(w+1)/2, & w \in [3 : \delta-1], \\ nw/2, & w=\delta-1 > 2. \end{cases}$	$\leq (1+\epsilon)d \frac{\ell}{w}$	$s < r$
New vector LRC in Construction 2	$\frac{\mu r-s}{\mu(r+\delta-1)}$	w^m	$\geq \mu w m^{\lceil \frac{r+\delta-1}{wm} \rceil} + 1$	$\leq (1+\epsilon)d \frac{\ell}{w}$	$s < r$
New vector LRC in Construction 3	$\frac{\mu r-s}{\mu(r+\delta-1)}$	2	$\geq \mu(r+\delta-1) + 2$	$\frac{3r+\delta-1}{2}$	$s < r$

vi) The new vector LRCs in Constructions 1 and 2 provide a flexible tradeoff between the sub-packetization level and repair bandwidth. Specifically, the repair bandwidth of the new vector LRCs in Constructions 1 and 2 is $1 + \epsilon$ times that of the constructions in [27], [28], and [38] under the same parameters n, k , and d , where $\epsilon = \frac{(\frac{r+\delta-1}{m}-1)(w-1)}{d}$ and can be controlled by varying m .

VII. CONCLUSION

In this paper, we proposed three constructions of optimal vector LRCs with small repair bandwidth, the required field sizes are comparable to the code lengths, and their sub-packetization levels are much smaller than that of existing vector LRCs. The first two new vector LRCs can provide a flexible tradeoff between the sub-packetization level and the repair bandwidth. The sub-packetization level of the third vector LRC is only 2.

ACKNOWLEDGMENT

The authors would like to thank the Associate Editor Prof. Viveck Cadambe and the two anonymous reviewers for their valuable suggestions and comments, which have greatly improved the presentation and quality of this paper.

REFERENCES

- [1] B. Calder, J. Wang, A. Ogus, N. Nilakantan, A. Skjolsvold, S. McKelvie, Y. Xu, S. Srivastav, J. Wu, H. Simitci, *et al.*, "Windows azure storage: a highly available cloud storage service with strong consistency," in *Proceedings of the Twenty-Third ACM Symposium on Operating Systems Principles*, pp. 143–157, 2011.
- [2] D. Borthakur, "The hadoop distributed file system: Architecture and design," *Hadoop Project Website*, vol. 11, no. 2007, p. 21, 2007.
- [3] D. Borthakur, R. Schmit, R. Vadali, S. Chen, and P. Kling, "HDFS RAID," <http://www.slideshare.net/ydn/hdfs-raid-facebook>, Nov. 20 2012.
- [4] A. G. Dimakis, P. B. Godfrey, Y. Wu, M. J. Wainwright, and K. Ramchandran, "Network coding for distributed storage systems," *IEEE transactions on information theory*, vol. 56, no. 9, pp. 4539–4551, 2010.
- [5] P. Gopalan, C. Huang, H. Simitci, and S. Yekhanin, "On the locality of codeword symbols," *IEEE Transactions on Information theory*, vol. 58, no. 11, pp. 6925–6934, 2012.
- [6] K. V. Rashmi, N. B. Shah, and P. V. Kumar, "Optimal exact-regenerating codes for distributed storage at the MSR and MBR points via a product-matrix construction," *IEEE Transactions on Information Theory*, vol. 57, no. 8, pp. 5227–5239, 2011.
- [7] I. Tamo, Z. Wang, and J. Bruck, "Zigzag codes: MDS array codes with optimal rebuilding," *IEEE Transactions on Information Theory*, vol. 59, no. 3, pp. 1597–1616, 2012.
- [8] D. S. Papailiopoulos, A. G. Dimakis, and V. R. Cadambe, "Repair optimal erasure codes through hadamard designs," *IEEE Transactions on Information Theory*, vol. 59, no. 5, pp. 3021–3037, 2013.
- [9] C. Tian, "Characterizing the rate region of the $(4, 3, 3)$ exact-repair regenerating codes," *IEEE Journal on Selected Areas in Communications*, vol. 32, no. 5, pp. 967–975, 2014.
- [10] Y. S. Han, H.-T. Pai, R. Zheng, and P. K. Varshney, "Update-efficient error-correcting product-matrix codes," *IEEE Transactions on Communications*, vol. 63, no. 6, pp. 1925–1938, 2015.
- [11] J. Li, X. Tang, and U. Parampalli, "A framework of constructions of minimal storage regenerating codes with the optimal access/update property," *IEEE Transactions on Information theory*, vol. 61, no. 4, pp. 1920–1932, 2015.
- [12] B. Sasidharan, M. Vajha, and P. V. Kumar, "An explicit, coupled-layer construction of a high-rate MSR code with low sub-packetization level, small field size and all-node repair," *arXiv preprint arXiv:1607.07335*, 2016.
- [13] Z. Wang, I. Tamo, and J. Bruck, "Explicit minimum storage regenerating codes," *IEEE Transactions on Information Theory*, vol. 62, no. 8, pp. 4466–4480, 2016.
- [14] J. Li and X. Tang, "Optimal exact repair strategy for the parity nodes of the $(k+2, k)$ zigzag code," *IEEE Transactions on Information Theory*, vol. 62, no. 9, pp. 4848–4856, 2016.
- [15] M. Ye and A. Barg, "Explicit constructions of high-rate MDS array codes with optimal repair bandwidth," *IEEE Transactions on Information Theory*, vol. 63, no. 4, pp. 2001–2014, 2017.

- [16] M. Ye and A. Barg, "Explicit constructions of optimal-access MDS codes with nearly optimal sub-packetization," *IEEE Transactions on Information Theory*, vol. 63, no. 10, pp. 6307–6317, 2017.
- [17] J. Li, X. Tang, and C. Tian, "A generic transformation for optimal repair bandwidth and rebuilding access in MDS codes," in *2017 IEEE International Symposium on Information Theory (ISIT)*, pp. 1623–1627, IEEE, 2017.
- [18] S. Goparaju, A. Fazeli, and A. Vardy, "Minimum storage regenerating codes for all parameters," *IEEE Transactions on Information Theory*, vol. 63, no. 10, pp. 6318–6328, 2017.
- [19] J. Li, X. Tang, and C. Tian, "A generic transformation to enable optimal repair in MDS codes for distributed storage systems," *IEEE Transactions on Information Theory*, vol. 64, no. 9, pp. 6257–6267, 2018.
- [20] S. Balaji, M. N. Krishnan, M. Vajha, V. Ramkumar, B. Sasidharan, and P. V. Kumar, "Erasure coding for distributed storage: An overview," *Science China Information Sciences*, vol. 61, no. 10, pp. 1–45, 2018.
- [21] Z. Chen and A. Barg, "Explicit constructions of MSR codes for clustered distributed storage: The rack-aware storage model," *IEEE Transactions on Information Theory*, vol. 66, no. 2, pp. 886–899, 2019.
- [22] H. Hou and P. P. Lee, "Binary MDS array codes with optimal repair," *IEEE Transactions on Information Theory*, vol. 66, no. 3, pp. 1405–1422, 2019.
- [23] M. Elyasi and S. Mohajer, "Cascade codes for distributed storage systems," *IEEE Transactions on Information Theory*, vol. 66, no. 12, pp. 7490–7527, 2020.
- [24] Y. Liu, J. Li, and X. Tang, "A generic transformation to generate MDS array codes with δ -optimal access property," *IEEE Transactions on Communications*, vol. 70, no. 2, pp. 759–768, 2022.
- [25] D. S. Papailiopoulos and A. G. Dimakis, "Locally repairable codes," *IEEE Transactions on Information Theory*, vol. 60, no. 10, pp. 5843–5855, 2014.
- [26] N. Prakash, G. M. Kamath, V. Lalitha, and P. V. Kumar, "Optimal linear codes with a local-error-correction property," in *2012 IEEE International Symposium on Information Theory Proceedings*, pp. 2776–2780, IEEE, 2012.
- [27] G. M. Kamath, N. Prakash, V. Lalitha, and P. V. Kumar, "Codes with local regeneration and erasure correction," *IEEE Transactions on Information Theory*, vol. 60, no. 8, pp. 4637–4660, 2014.
- [28] A. S. Rawat, O. O. Koyluoglu, N. Silberstein, and S. Vishwanath, "Optimal locally repairable and secure codes for distributed storage systems," *IEEE Transactions on Information Theory*, vol. 60, no. 1, pp. 212–236, 2014.
- [29] I. Tamo and A. Barg, "A family of optimal locally recoverable codes," *IEEE Transactions on Information Theory*, vol. 60, no. 8, pp. 4661–4676, 2014.
- [30] A. Wang and Z. Zhang, "Repair locality with multiple erasure tolerance," *IEEE Transactions on Information Theory*, vol. 60, no. 11, pp. 6979–6987, 2014.
- [31] H. Cai, M. Cheng, C. Fan, and X. Tang, "Optimal locally repairable systematic codes based on packings," *IEEE Transactions on Communications*, vol. 67, no. 1, pp. 39–49, 2018.
- [32] Y. Luo, C. Xing, and C. Yuan, "Optimal locally repairable codes of distance 3 and 4 via cyclic codes," *IEEE Transactions on Information Theory*, vol. 65, no. 2, pp. 1048–1053, 2018.
- [33] V. Guruswami, C. Xing, and C. Yuan, "How long can optimal locally repairable codes be?," *IEEE Transactions on Information Theory*, vol. 65, no. 6, pp. 3662–3670, 2019.
- [34] L. Jin, "Explicit construction of optimal locally recoverable codes of distance 5 and 6 via binary constant weight codes," *IEEE Transactions on Information Theory*, vol. 65, no. 8, pp. 4658–4663, 2019.
- [35] B. Chen, W. Fang, S.-T. Xia, J. Hao, and F.-W. Fu, "Improved bounds and singleton-optimal constructions of locally repairable codes with minimum distance 5 and 6," *IEEE Transactions on Information Theory*, vol. 67, no. 1, pp. 217–231, 2020.
- [36] H. Cai, Y. Miao, M. Schwartz, and X. Tang, "On optimal locally repairable codes with super-linear length," *IEEE Transactions on Information Theory*, vol. 66, no. 8, pp. 4853–4868, 2020.
- [37] H. Cai and M. Schwartz, "On optimal locally repairable codes and generalized sector-disk codes," *IEEE Transactions on Information Theory*, vol. 67, no. 2, pp. 686–704, 2021.
- [38] M. N. Krishnan, A. N. R., and P. V. Kumar, "Codes with combined locality and regeneration having optimal rate, $d_{\min}$ and linear field size," in *2018 IEEE International Symposium on Information Theory (ISIT)*, pp. 1196–1200, IEEE, 2018.
- [39] M. N. Krishnan, V. Lalitha, and S. Didugu, "Codes with minimum bandwidth cooperative local regeneration," *IEEE Communications Letters*, vol. 24, no. 12, pp. 2669–2673, 2020.
- [40] O. Alrabiah and V. Guruswami, "An exponential lower bound on the sub-packetization of MSR codes," in *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing*, pp. 979–985, 2019.
- [41] J. Li, X. Tang, H. Hou, Y. S. Han, B. Bai, and G. Zhang, "PMDS array codes with small sub-packetization, small repair bandwidth/rebuilding access," *IEEE Transactions on Information Theory*, vol. 69, no. 3, pp. 1551–1566, 2023.
- [42] L. Holzbaur, S. Puchinger, E. Yaakobi, and A. Wachter-Zeh, "Partial MDS codes with regeneration," *IEEE Transactions on Information Theory*, vol. 67, no. 10, pp. 6425–6441, 2021.
- [43] J. Li, Y. Liu, X. Tang, Y. S. Han, B. Bai, and G. Zhang, "MDS array codes with (near) optimal repair bandwidth for all admissible repair degrees," *IEEE Transactions on Communications*, vol. 71, no. 10, pp. 5633–5646, 2023.
- [44] J. Li, Y. Liu, and X. Tang, "A systematic construction of MDS codes with small sub-packetization level and near-optimal repair bandwidth," *IEEE Transactions on Information Theory*, vol. 67, no. 4, pp. 2162–2180, 2021.
- [45] J. Li, Y. Liu, X. Tang, Y. S. Han, B. Bai, and G. Zhang, "MDS array codes with small sub-packetization levels and small repair degrees," *arXiv preprint arXiv: 2308.14527*, 2023.



Jie Li (Senior Member, IEEE) received the B.S. and M.S. degrees in mathematics from Hubei University, Wuhan, China, in 2009 and 2012, respectively, and received the Ph.D. degree from the department of communication engineering, Southwest Jiaotong University, Chengdu, China, in 2017.

From 2015 to 2016, he was a visiting Ph.D. student at the Department of Electrical Engineering and Computer Science, The University of Tennessee at Knoxville, TN, USA. From 2017 to 2019, he was a postdoctoral researcher at the Department of Mathematics, Hubei University, Wuhan, China. From 2019 to 2021, he was a postdoctoral researcher at the Department of Mathematics and Systems Analysis, Aalto University, Finland. He is currently a Senior Researcher with the Theory Lab, Central Research Institute, 2012 Labs, Huawei Technologies Co., Ltd., Hong Kong SAR, China. His research interests include coding for distributed storage and distributed computing, private information retrieval, and sequence design.

Dr. Li received the IEEE Jack Keil Wolf ISIT Student Paper Award and the Honor of Outstanding Graduates of Sichuan Province both in 2017.



Han Cai (Member, IEEE) received the B.S. and M.S. degrees in mathematics from Hubei University, Wuhan, China, in 2009 and 2013, respectively, and the Ph.D. degree from the Department of Communication Engineering, Southwest Jiaotong University, Chengdu, China, in 2017. From October 2015 to October 2017, he was a Visiting Ph.D. Student with the Faculty of Engineering, Information and Systems, University of Tsukuba, Japan. From 2018 to 2021, he was a Post-Doctoral Fellow at the School of Electrical and Computer Engineering, Ben-Gurion University of the Negev, Israel. In 2021, he joined Southwest Jiaotong University, where he currently holds a tenure-track position. His research interests include coding theory and sequence design.



Xiaohu Tang (Senior Member, IEEE) received the B.S. degree in applied mathematics from the Northwest Polytechnic University, Xi'an, China, the M.S. degree in applied mathematics from the Sichuan University, Chengdu, China, and the Ph.D. degree in electronic engineering from the Southwest Jiaotong University, Chengdu, China, in 1992, 1995, and 2001 respectively.

From 2003 to 2004, he was a research associate in the Department of Electrical and Electronic Engineering, Hong Kong University of Science and Technology. From 2007 to 2008, he was a visiting professor at University of Ulm, Germany. Since 2001, he has been in the School of Information Science and Technology, Southwest Jiaotong University, where he is currently a professor. His research interests include coding theory, network security, distributed storage and information processing for big data.

Dr. Tang was the recipient of the National excellent Doctoral Dissertation award in 2003 (China), the Humboldt Research Fellowship in 2007 (Germany), and the Outstanding Young Scientist Award by NSFC in 2013 (China). He served as Associate Editors for several journals including *IEEE Transactions on Information Theory* and *IEICE Transactions on Fundamentals*, and served on a number of technical program committees of conferences.



Yunghsiang S. Han (S'90-M'93-SM'08-F'11) was born in Taipei, Taiwan, in 1962. He received B.Sc. and M.Sc. degrees in electrical engineering from the National Tsing Hua University, Hsinchu, Taiwan, in 1984 and 1986, respectively, and a Ph.D. from the School of Computer and Information Science, Syracuse University, Syracuse, NY, in 1993. From 1986 to 1988, he was a lecturer at Ming-Hsin Engineering College, Hsinchu, Taiwan. He was a teaching assistant from 1989 to 1992 and a research associate in the School of Computer and Information

Science at Syracuse University from 1992 to 1993. From 1993 to 1997, he was an Associate Professor in the Department of Electronic Engineering at Hua Fan College of Humanities and Technology, Taipei Hsien, Taiwan. He was with the Department of Computer Science and Information Engineering at National Chi Nan University, Nantou, Taiwan from 1997 to 2004. He was promoted to Professor in 1998. He was a visiting scholar in the Department of Electrical Engineering at the University of Hawaii at Manoa, HI from June to October 2001, the SUPRIA visiting research scholar in the Department of Electrical Engineering and Computer Science and CASE center at Syracuse University, NY from September 2002 to January 2004 and July 2012 to June 2013, and the visiting scholar in the Department of Electrical and Computer Engineering at the University of Texas at Austin, TX from August 2008 to June 2009. He was with the Graduate Institute of Communication Engineering at National Taipei University, Taipei, Taiwan from August 2004 to July 2010. From August 2010 to January 2017, he was Chair Professor with the Department of Electrical Engineering at the National Taiwan University of Science and Technology. From February 2017 to February 2021, he was with the School of Electrical Engineering & Intelligentization at Dongguan University of Technology, China. Now he is with the Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China. He is also a Chair Professor at National Taipei University since February 2015. His research interests are in error-control coding, wireless networks, and security.

Dr. Han was a winner of the 1994 Syracuse University Doctoral Prize and a Fellow of IEEE. One of his papers won the prestigious 2013 ACM CCS Test-of-Time Award in cybersecurity.



Bo Bai (Senior Member, IEEE) received the B.S. degree (Hons.) from the School of Communication Engineering, Xidian University, Xian, China, in 2004, and the Ph.D. degree from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2010.

He was a Research Assistant and a Research Associate with the Department of Electronic and Computer Engineering, The Hong Kong University of Science and Technology, from April 2009 to September 2010 and October 2010 to April 2012, respectively. From July 2012 to January 2017, he was an Assistant Professor with the Department of Electronic Engineering, Tsinghua University. He has obtained the support from the Backbone Talents Supporting Project of Tsinghua University. Currently, he is an Information Theory Scientist and the Director of the Theory Laboratory, 2012 Labs, Huawei Technologies Company Ltd., Hong Kong. He has authored more than 130 papers in major IEEE and ACM journals and conferences, two book chapters, and one textbook. His research interests include semantic information theory, B5G/6G networking, and graph informatics. He served as a member for the IEEE ComSoc WTC Committee and the IEEE ComSoc SPCE Committee. He also served as a TPC Member for several IEEE conferences, such as ICC, Globecom, WCNC, VTC, and ICC. He was a recipient of the Student Travel Grant at IEEE Globecom 2009 and the Best Paper Award at the IEEE ICC 2016. He was invited as a Young Scientist Speaker at the IEEE TTM 2011. He received the Honor of Outstanding Graduates of Shaanxi Province and the Honor of Young Academic Talent of Electronic Engineering in Tsinghua University. He is one of the Founding Vice Chair of the IEEE TCCN SIG on Social Behavior Driven Cognitive Radio Networks. He served as the TPC Co-Chair for the IEEE Infocom 2018 1st AoI Workshop and the IEEE Infocom 2019 2nd AoI Workshop, the TPC Co-Chair for the IEEE ICC 2018, and an Industrial Forum and the Exhibition Co-Chair for the IEEE HotICN 2018. He also served as an Associate Editor of *IEEE COMMUNICATIONS SURVEYS AND TUTORIALS* and a reviewer for several major IEEE and ACM journals and conferences.



Gong Zhang is a chief researcher scientist in future network architecture of Huawei 2012 Labs. He has over 20 years of research experience as a system architect in Networks, distributed systems, and communication systems. He had contributed more than 90 patents globally. He was a Senior Researcher, leading future internet research and co-operative communication, and did Mobility research program since 2005. In 2009, Principal Researcher, in charge of advanced network technology research department, leading research on future networks,

distributed computing, database system, and data analysis. He proposed a stream-based research system in the network to maintain the network and find added value for carriers. He was a Principal Researcher, leading system group in data mining and machine learning since 2012. Recently, the major research directions are Network architecture and large-scale distributed systems.